

KARNATAKA STATE OPEN UNIVERISTY

A project report entitled.

“Agromart”

A project Report submitted in partial fulfillment of the award of the degree of

MASTER OF COMPUTER APPLICATION

SUBMITTED BY

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May - 2026

KARNATAKA STATE OPEN UNIVERISTY

Certificate

This is to certify that the project work entitled “**Agromart**” is a Bonafide work carried out by **MOKSHITH P J (Roll no: 08P232036900049)** in partial fulfillment of a project in **Master of Science in Computer Science, under Karnataka State Open University, Mysuru, during the year May 2026.** The project report has been approved as it satisfies the academic requirements in respect to project work prescribed for the Master of Science in Computer Science.

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ACKNOWLEDGEMENT

We would like to sincerely thank all the people who were directly or indirectly involved in this project, without whose help the completion of this project would have been impossible.

We thank **Mrs. Suneetha**, Associate Professor, Department of Computer Science, for his valuable guidance and the extra effort that he is making every time to ensure correctness and perfection of what we are doing.

Our special thanks to all friends who have provided timely help and genuine feedback which helped us improve a lot.

Finally, we thank our parents, who financially supported us throughout the project and provided the support that we need throughout our life.

Place: Bangalore

By,

Date: 12-05-2026

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Chapter 1

INTRODUCTION

1.1. Introduction of the System.

Agromart is a web-based project that aims to help Farmers and Consumers. The main users of the system, the Farmers usually face a lot of problems in finding the good fertilizers, quality equipment's or any other agriculture related items that will be helpful for them. The proposed system Agromart will solve the problem by developing a system that will provide all the agriculture-related information in one place.

1.1.1. Project Title Agromart

1.1.2. Category This application is a Client Defined Web Application. This application is developed with the help of PHP and MySQL Server.

1.1.3. Overview The proposed framework is a Web based program that maintains a consolidated archive of all related information. The system allows you to quickly access the relevant details. Farmers can view and purchase the required tools or items from the website and also upload crops they farmed. Vendors who want to purchase crops from farmers bid for the crops.

1.1.4. Background The Agromart project sounds like it could be an initiative related to agriculture and market development. Agriculture is a vital sector in many economies, providing food, raw materials, and employment opportunities. The ultimate goal would likely be to enhance the overall productivity, profitability, and resilience of the agricultural sector while improving the livelihoods of farmers and ensuring food security for the population.

1.2. Objectives of the System

- The Objective of the proposed system is to provide a platform where both users, Farmers and Consumers will get a system that will helpful for them regarding their respected problems.
- Farmers will get the items such as Fertilizers and any other Agriculture related items that are needed for them.
- They can also upload the Products that they are willing to sell.
- In other way Consumers will upload the items that they are selling and also will see the Products that are uploaded by Farmers and bid for them.

1.3. Importance and Scope of the System.

The main scope of the project is to provide a website for the betterment of Farmers. Now farmers facing problems like they are not getting the expected price for their crops and also they are not able to purchase the necessary products because of the mediators.

1.4. Structure of the System

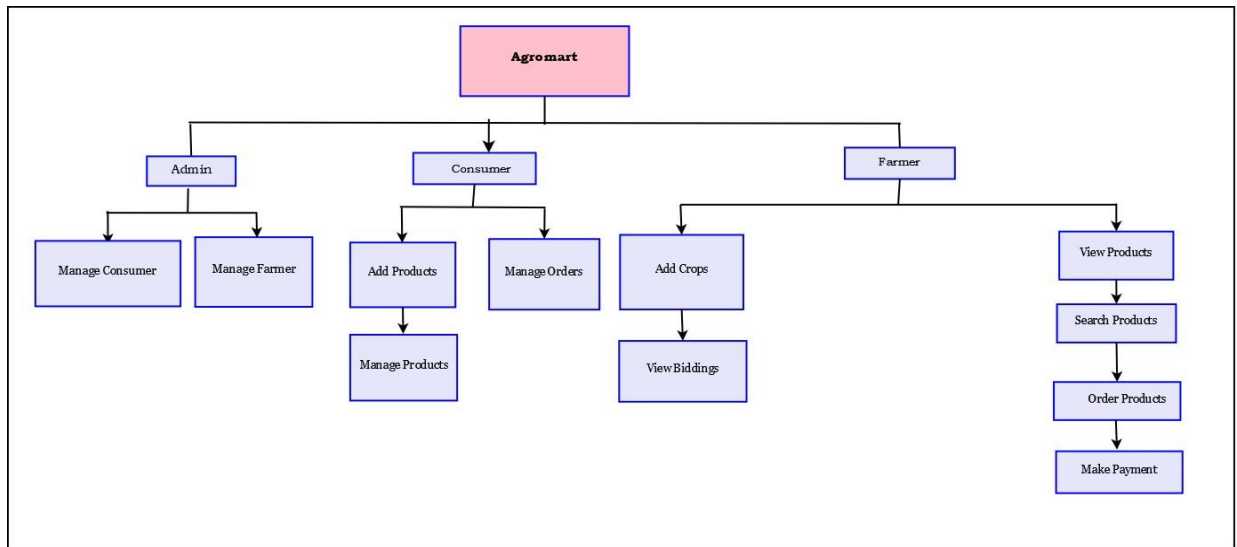


Figure 1.1 Structure of the System

The system Agromart contains three modules namely Admin, Consumer, farmer. The Admin manages the whole system. The consumer will add the services like adding products, Managing Orders, Managing Products. The Farmers can perform the activities like viewing Products, search for products, order Products and also make payment.

1.5. System Architecture

Software architecture is the set of structures needed to reason about a software system and the discipline of creating such structures and systems. Each structure comprises software elements, relations among them, and properties of both elements and relations.

The architecture of a software system is a metaphor, analogous to the architecture of a building. It functions as the blueprints for the system and the development project, which project management can later use to extrapolate the tasks necessary to be executed by the teams and people involved.

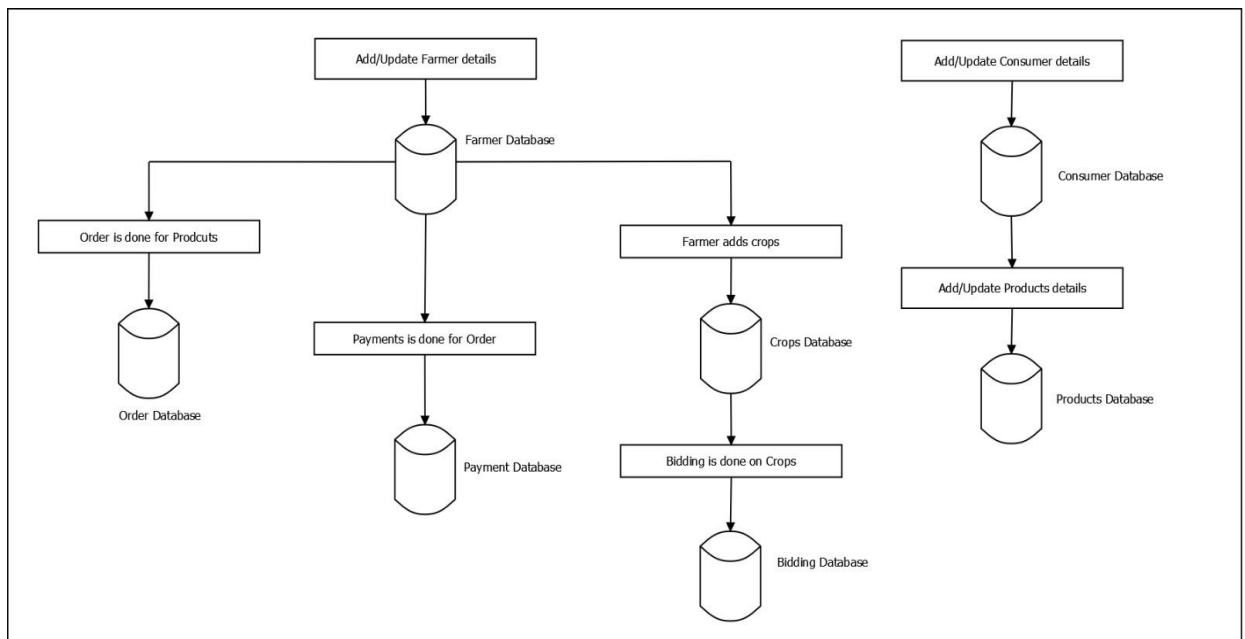


Figure 2.2 Structure of the System

1.6. Software/Hardware used for the development

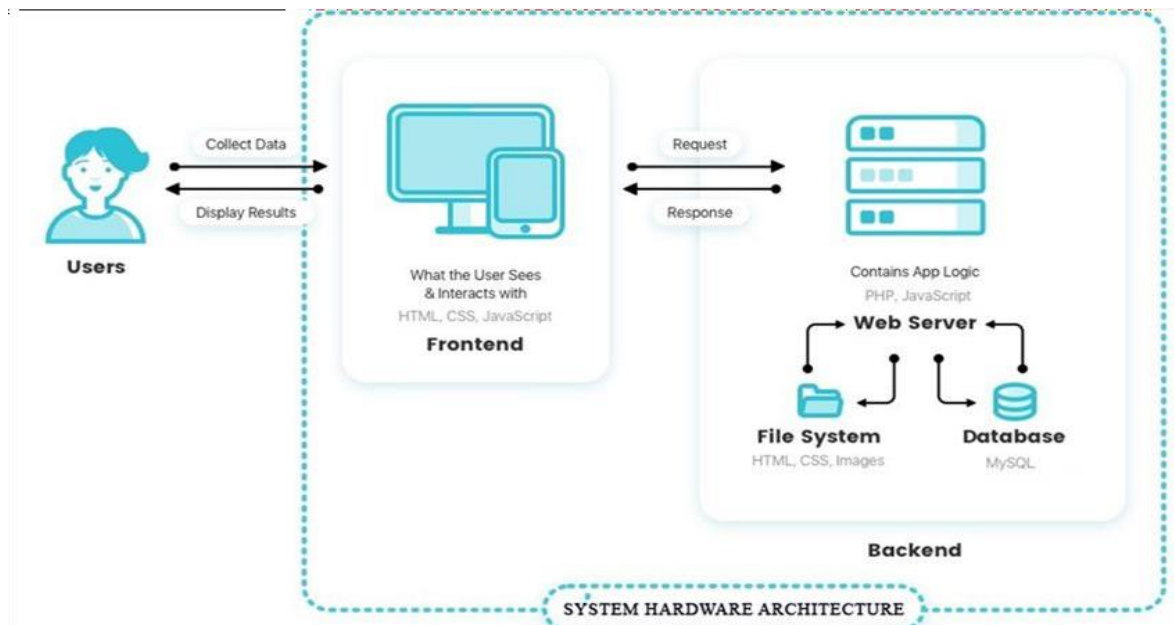


Figure 1.3 System Hardware Architecture

This Web Application is developed using free and open-source cross-platform web server solution stack for Windows WAMP. Front end application is developed using html, CSS, Javascript programming language. The data relieved by user interfaces are stored in the MySQL Server passing through Firewall. Apache Web Server will function as Web Server Software in the background. The application will be hosted on the Web using third-party service providers, where they implement Background Server for the backup purpose. Users may interact with the System using Workstation, Laptop or Smart phones.

1.7. Software/Hardware required for the implementation.

1.7.1. Software Requirements (for development)

IDE: Sublime Text 3

Server: Apache 2.4.39

Database: MySQL 5.7.26

Client Side: Chrome, Opera, Mozilla Firefox

1.7.2. Hardware Requirements (for development)

Operating System: Windows 7, Linux, Mac OS

RAM: 4GB or above

HDD: 500GB or above

Processor: 2.4 GHz or above

1.7.3. Software Requirements (for implementation)

IDE: Sublime Text 3

Server: Apache 2.4.39

Database: MySQL 5.7.26

Client Side: Chrome, Opera, Mozilla Firefox

1.7.4. Hardware Requirements (for implementation)

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HDD: 500GB or above

Processor: 2.4 GHz or above

➤ HTML

Hyper Text Markup Language (HTML) is a simple markup system used to create hypertext documents that are portable from one platform to another. HTML documents are SGML documents with generic semantics that are appropriate for representing information from a wide range of applications. HTML markup can represent hypertext news, mail, documentation, and hypermedia; menus of options; database query results; simple structured documents with in-lined graphics; and hypertext views of existing bodies of information. HTML has been in use by the World-Wide Web (WWW) global information initiative since 1990. The HTML 3.0 specification provides a number of new features, and is broadly backwards compatible with HTML 2.0. It is defined as an application of International Standard ISO ISO8879:1986 Standard Generalized Markup Language (SGML).

➤ CSS

Cascading Style Sheets is a style sheet language used for describing the presentation of a document written in a markup language such as HTML. CSS is a cornerstone technology of the World Wide Web, alongside HTML and JavaScript. It is designed to enable the separation of presentation and content, including layout, colors, and fonts. It enables you to do this independent of the HTML that makes up each web page.

➤ PHP

PHP is a recursive acronym for "PHP: Hypertext Pre-processor". PHP is a server- side scripting language that is embedded in HTML. It is used to manage dynamic content, databases, session tracking, even build entire e-commerce sites. It is integrated with a number of popular databases, including MySQL, PostgreSQL, Oracle, Sybase, Informix, and Microsoft SQL Server. PHP is pleasingly zippy in its execution, especially when compiled as an Apache module on the Unix side. The MySQL server, once started, executes even very complex queries with huge result sets in record-setting time. PHP supports a large number of major protocols such as POP3, IMAP, and LDAP. PHP4 added support for Java and distributed object architectures (COM and CORBA), making n-tier development a possibility for the first time. PHP is forgiving: PHP language tries to be as forgiving as possible Syntax is C-Like. PHP performs system functions, i.e. from files on a system it can create, open, read, write, and close them. PHP can handle forms, i.e. gather data from files, save data to a file, through email you can send data, return data to the user. You add, delete, modify elements within your database through PHP.

➤ My SQL

MySQL is the world's most widely used opensource Relational Database Management System (RDBMS) that runs as a server providing multi-user access to a number of databases. The MySQL development project has made its source code available under the terms of the GNU General Public License, as well as under a variety of proprietary agreements. MySQL was owned and sponsored by a single for profit firm, the Swedish company MySQL AB, now owned by Oracle Corporation. MySQL is a popular choice of database for use in web applications, and is a central development of the widely used LAMP opensource web application software stack. LAMP is an acronym for "Linux, Apache, MySQL, and Perl/PHP/Python" Free software-opensource projects that require a full feature database management system often use MySQL.

➤ Microsoft visual studio

Visual Studio Code is a lightweight but powerful source code editor which runs on your desktop and is available for Windows, macOS and Linux. It comes with built-in support for JavaScript, TypeScript and Node.js and has a rich ecosystem of extensions for other languages and runtimes (such as C++, C#, Java, Python, PHP, Go, .NET).

Chapter 2

METHODOLOGY

2.1. System Design

Software design is a process by which the software requirements are translated into presentation of software components, interfaces, and data necessary for the implementation phase. The System Design Document describes the system requirements, operating system environment, subsystem architecture, files and database design, input formats, output layouts, human-machine interfaces, detailed design, processing logic, and external interfaces.

2.1.1. Functional Decomposition

a. System Software Architecture

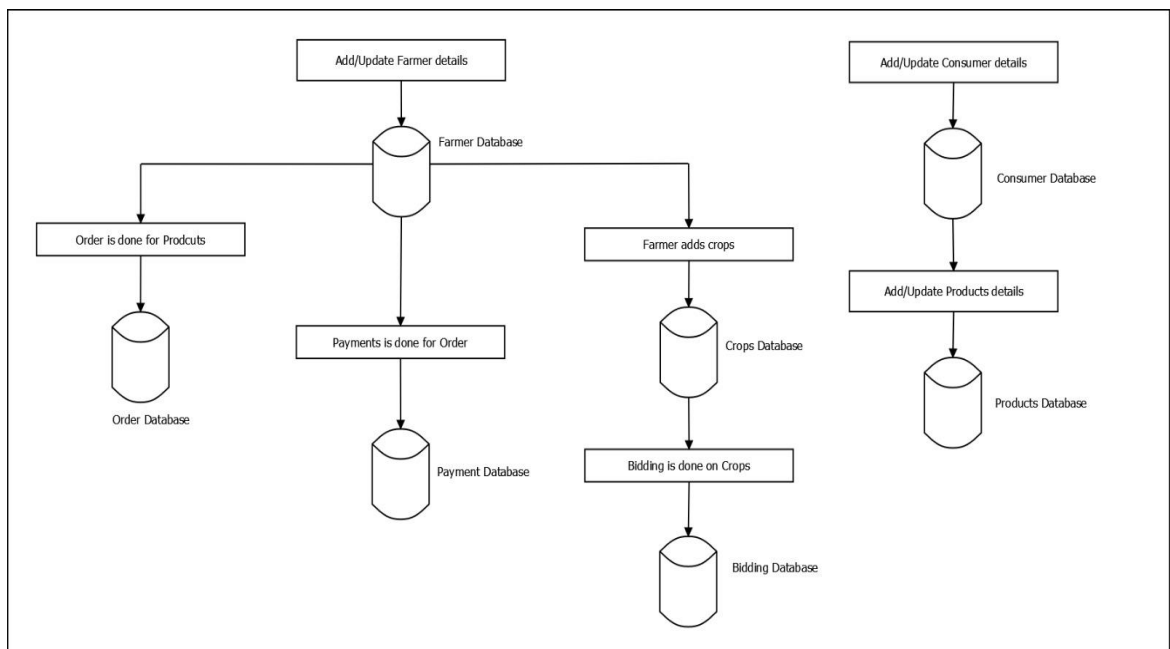


Figure 2.1 System software Architecture

In this System Software Architecture, we can find the working of the software. The user will be added by the admin. Then through logging in the consumer can add or update the product details, bid the crops. The farmer can add the crops, they can view the payment.

b. System Technical Architecture

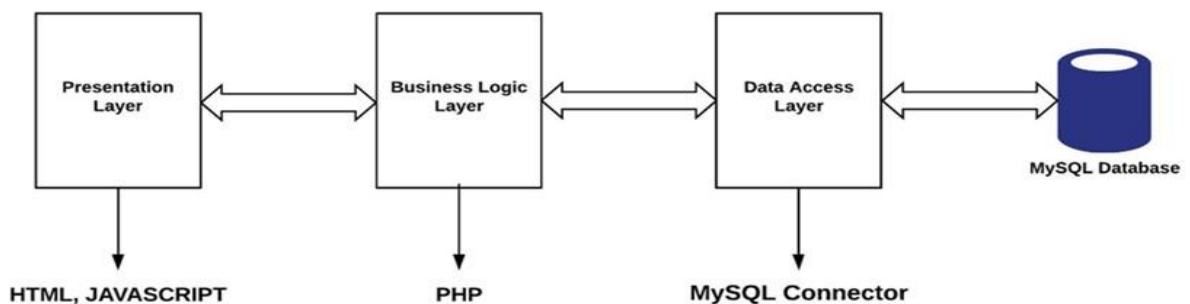


Figure 2.2 System Technical Architecture

In system technical architecture we discuss the technologies used in each layer. In presentation layer html, and Javascript are used. In business logic layer PHP is used. In data access layer MYSQL connector is used. MYSQL database technology is used for storing data.

c. System Hardware Architecture

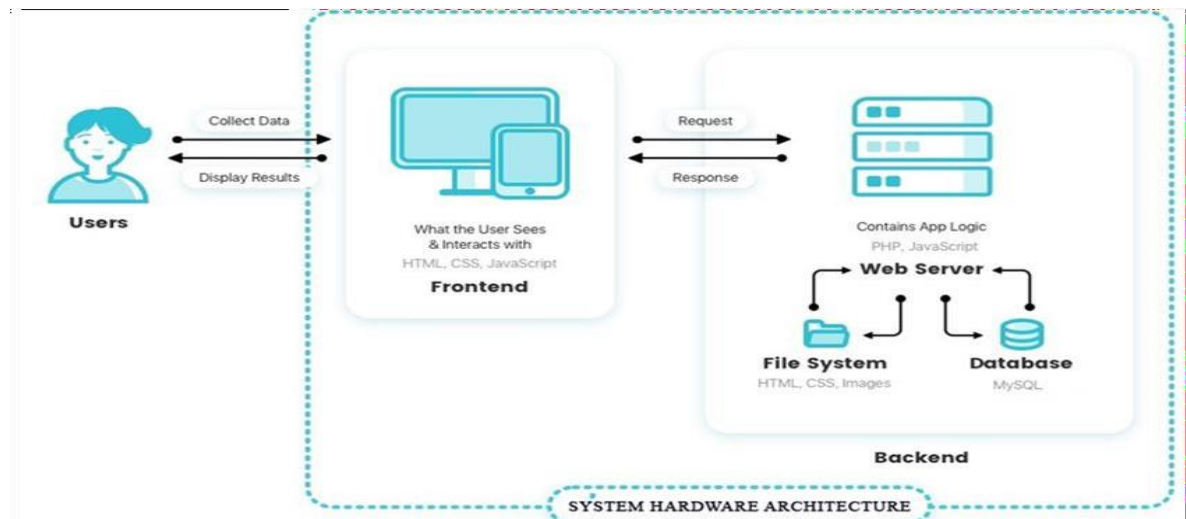


Figure 2.3 System Hardware Architecture

This Web Application is developed using free and open-source cross-platform web server solution stack for Windows WAMP. Front end application is developed using html, CSS, Javascript programming language. The data relieved by user interfaces are stored in the MySQL Server passing through Firewall. Apache Web Server will function as Web Server Software in the background. The application will be hosted on the Web using third-party service providers, where they implement Background Server for the backup purpose. Users may interact with the System using Workstation, Laptop or Smart phones.

d. External Interfaces

- Name of application: Agromart
- Owner of application:
- Details of interface: Login Panel, Register Panel, Consumer Panel, former Panel
- Type of interface: Graphical user interface

2.1.2. Description of Programs

a. Context Flow Diagram

Context Flow Diagram is a top level (also known as level0) data flow diagram. It contains only one process node that generalizes the functions of the entire system in relationship to external entities. In the context diagram the entire system is treated as a single process and all its inputs, outputs, sinks and sources are identified.

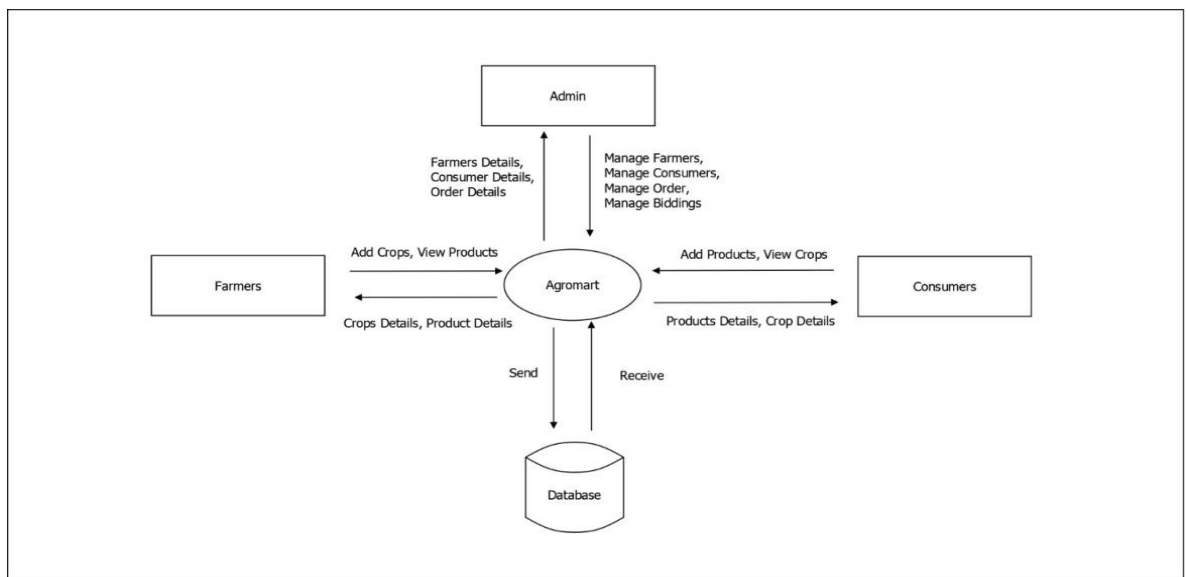


Figure 2.4 Context flow diagram

b. Data Flow Diagrams

Data Flow Diagram is a graphical representation of a system or a portion of the system. It consists of data flow, process, sources and sink and stores all the description through the use of easily understandable symbols. DFD is one of the most important modeling tools. It issued to model the system, components that interact with the system, and uses the data and information flows in the system. DFD shows the information moves through, and how it is modified by a series of transformations. It is a graphical representation that depicts information that moves from input to output. DFD is also known as bubble charts or Data Flows Graphs. DFD may be used to represent the system at any level of abstraction.

2.1.2.1. DFD (Level 1)

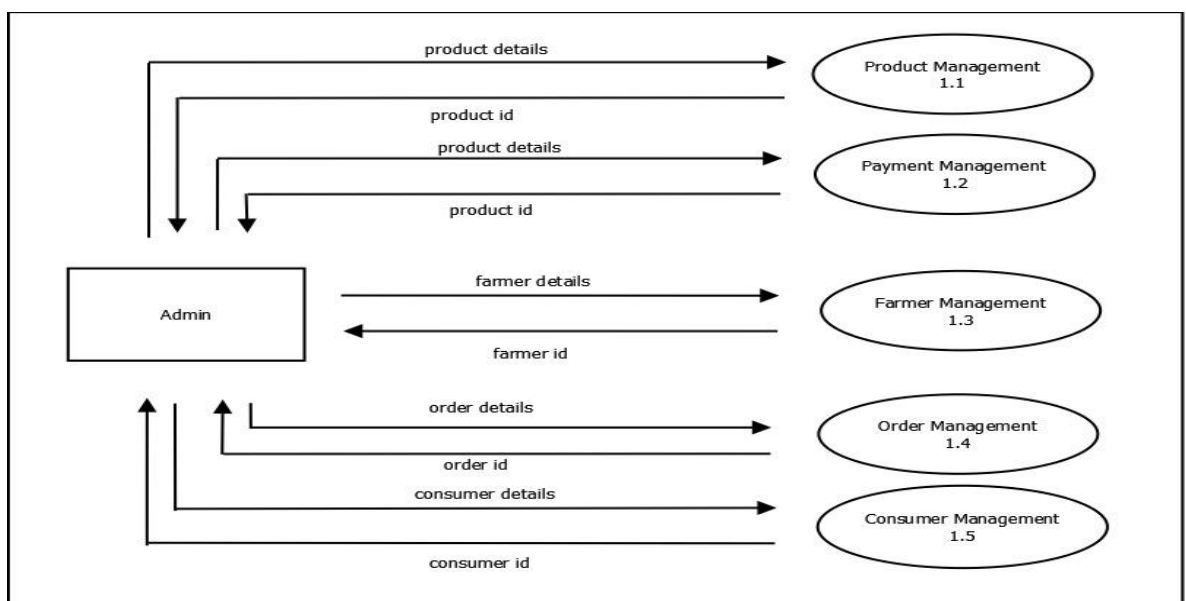


Figure 2.5 DFD Levels 1

2.1.2.2. DFD (Level 2)

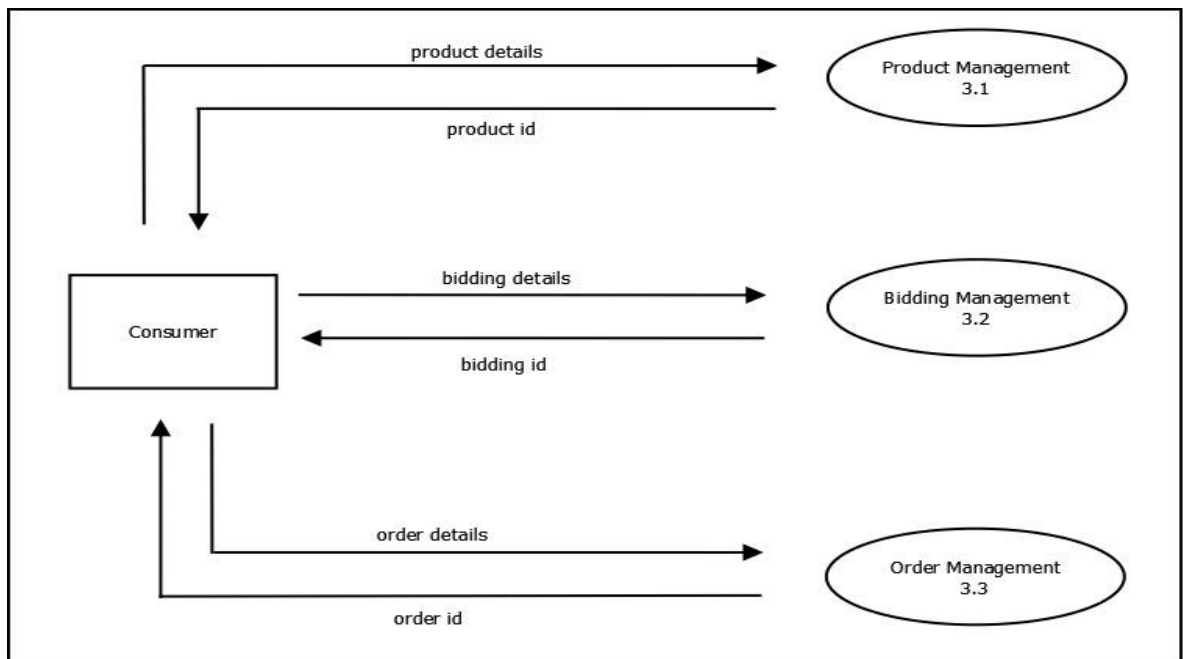


Figure 2.6 DFD Levels 2

2.1.1.3 DFD (Level 3)

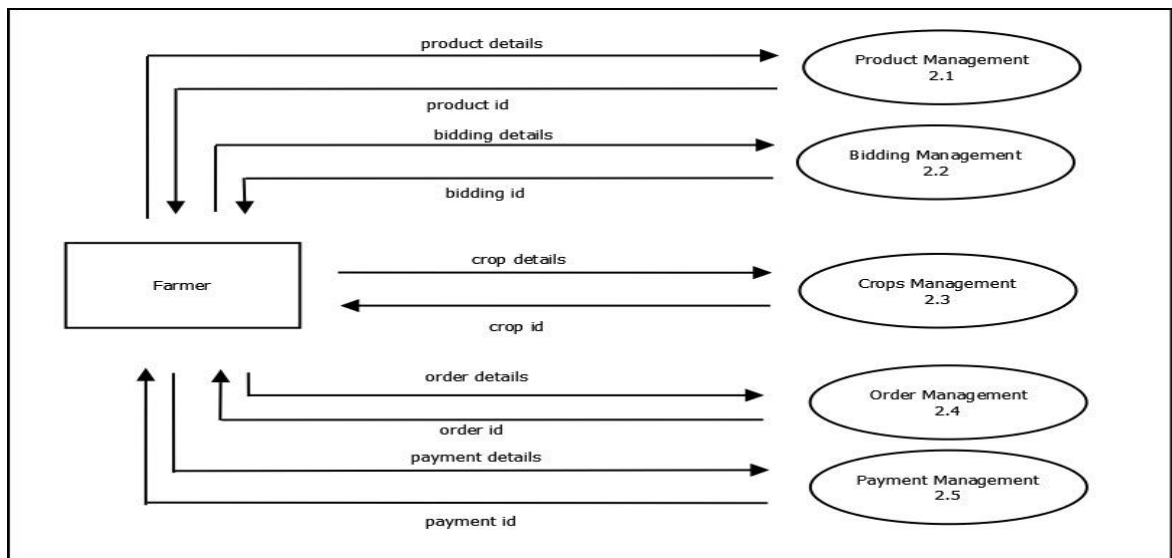


Figure 2.7 DFD Levels 3

2.2. Database Design

2.2.1. Database Identification

- To denote Database names, we use db.
- To name the tables we use small letters.
- To denote Primary key, we use pk.
- To denote foreign keys, we use fk.
- To name columns we use small letters

2.2.2. Table Definition

admin

Column	Type	Null	Description	Example
admin_id (<i>Primary</i>)	int(11)	No	To store Admin ID	1001
admin_email	varchar(30)	No	To store Admin Email	admin@gmail.com
admin_username	varchar(30)	No	To store Admin Username	admin
admin_password	varchar(30)	No	To store Admin Password	admin

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	admin_id	1	A	No	

consumer

Column	Type	Null	Description	Example
consumer_id (<i>Primary</i>)	int(11)	No	To store Consumer ID	2001
consumer_name	varchar(30)	No	To store Consumer Name	Ravi
consumer_email	varchar(30)	No	To store Consumer Email	ravi@gmail.com
consumer_contact_number	varchar(12)	No	To store Consumer Contact Number	9876543210
consumer_address	text	No	To store Consumer Address	Bangalore – 560012
consumer_username	varchar(30)	No	To store Consumer Username	Admin
consumer_password	varchar(30)	No	To store Consumer Password	admin

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	consumer_id	1	A	No	

crops

Column	Type	Null	Description	Example
crop_id (Primary)	int(11)	No	To store Crop ID	3001
crop_name	varchar(30)	No	To store Crop Name	Rice
crop_description	text	No	To store Crop Description	Sample
farmer_id	int(11)	No	To store Farmer ID	2001
crop_initial_amount	int(11)	No	To store Initial Amount	100
crop_amount	int(11)	Yes	To store Crop Price	500
consumer_id	int(11)	Yes	To store Consumer ID	6001
image_location	text	No	To store Crop Image location	./images/1.jpg

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	crop_id	1	A	No	

farmers

Column	Type	Null	Description	Example
farmer_id (Primary)	int(11)	No	To store Farmer ID	4005
farmer_name	varchar(30)	No	To store Farmer Name	Sample
farmer_email	varchar(30)	No	To store Farmer Email	sample@gmail.com
farmer_contact_number	varchar(12)	No	To store Farmer Contact Number	9876543210
farmer_address	text	No	To store Farmer Address	Bangalore - 560012
farmer_username	varchar(30)	No	To store Farmer Username	Admin
farmer_password	varchar(30)	No	To store Farmer Password	admin

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	farmer_id	2	A	No	

feedbacks

Column	Type	Null	Description	Example
feedback_id (Primary)	int(11)	No	To store Feedback ID	6415
feedback_from	varchar(30)	No	To store the person who sent Feedback	sample@gmail.com
feedback_to	varchar(30)	No	To store to whom the feedback has been sent	abc@gmail.com
feedback_subject	varchar(40)	No	To store Feedback Subject	Sample
feedback_message	text	No	To store Feedback Message	Sample

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	feedback_id	0	A	No	

notifications

Column	Type	Null	Description	Example
notification_id (<i>Primary</i>)	int(11)	No	To store Notification ID	4152
notification_from_id	int(11)	No	To store Notification From User ID	5412
notification_from	varchar(30)	No	To store Notification From Email ID	agfd@gmail.com
notification_to	varchar(30)	No	To store Notification to Email ID	jhf@gmail.com
notification_subject	varchar(30)	No	To store Notification Subject	Sample
notification_message	text	No	To store Notification Message	Sample

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	notification_id	2	A	No	

orders

Column	Type	Null	Description	Example
order_id (<i>Primary</i>)	int(11)	No	To store Order ID	7458
farmer_id	int(11)	No	To store Farmer ID	4125
product_id	int(11)	No	To store Product ID	4444
invoice_number	varchar(30)	No	To store Invoice Number	KJGUF8741
amount	int(11)	No	To store Order amount	4111
status	varchar(30)	No	To store Status of Order	ORDER PLACED
datetime	timestamp	No	To store datetime	05-06-2024 12:54 PM

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	order_id	1	A	No	

payments

Column	Type	Null	Description	Example
payment_id (<i>Primary</i>)	int(11)	No	To store Payment ID	5447
invoice_number	varchar(30)	No	To store Invoice Number	FUF8745
status	varchar(30)	No	To store Payment Status	PAYMENT DONE
datetime	timestamp	No	To store datetime	20-05-2024 01:65 PM
amount	int(11)	No	To store Amount	5412

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	payment_id	1	A	No	

products

Column	Type	Null	Description	Example
product_id (<i>Primary</i>)	int(11)	No	To store Product ID	5412
product_name	varchar(30)	No	To store Product Name	Sample
product_description	text	No	To store Product Description	Sample
product_amount	int(11)	No	To store Product Amount	54112
consumer_id	int(11)	No	To store Consumer ID	4125
image_location	text	No	To store Product Image Location	./images/abv.jpg

Indexes

Keyname	Type	Unique	Packed	Column	Cardinality	Collation	Null	Comment
PRIMARY	BTREE	Yes	No	product_id	1	A	No	

2.3.3. ER diagram

Entity relationship diagram is used in modern database software engineering to illustrate logical structure of database. It is a relational schema database modelling method used to model a system and approach. This approach is commonly used in database design. The diagram created using this method is called the E-R diagram.

An entity-relationship (ER) diagram is a specialized graphic that illustrates the relationships between entities in a database. ER diagrams often use symbols to represent three different types of information. Boxes are commonly used to represent entities. Diamonds are normally used to represent relationships and ovals are used to represent attributes.

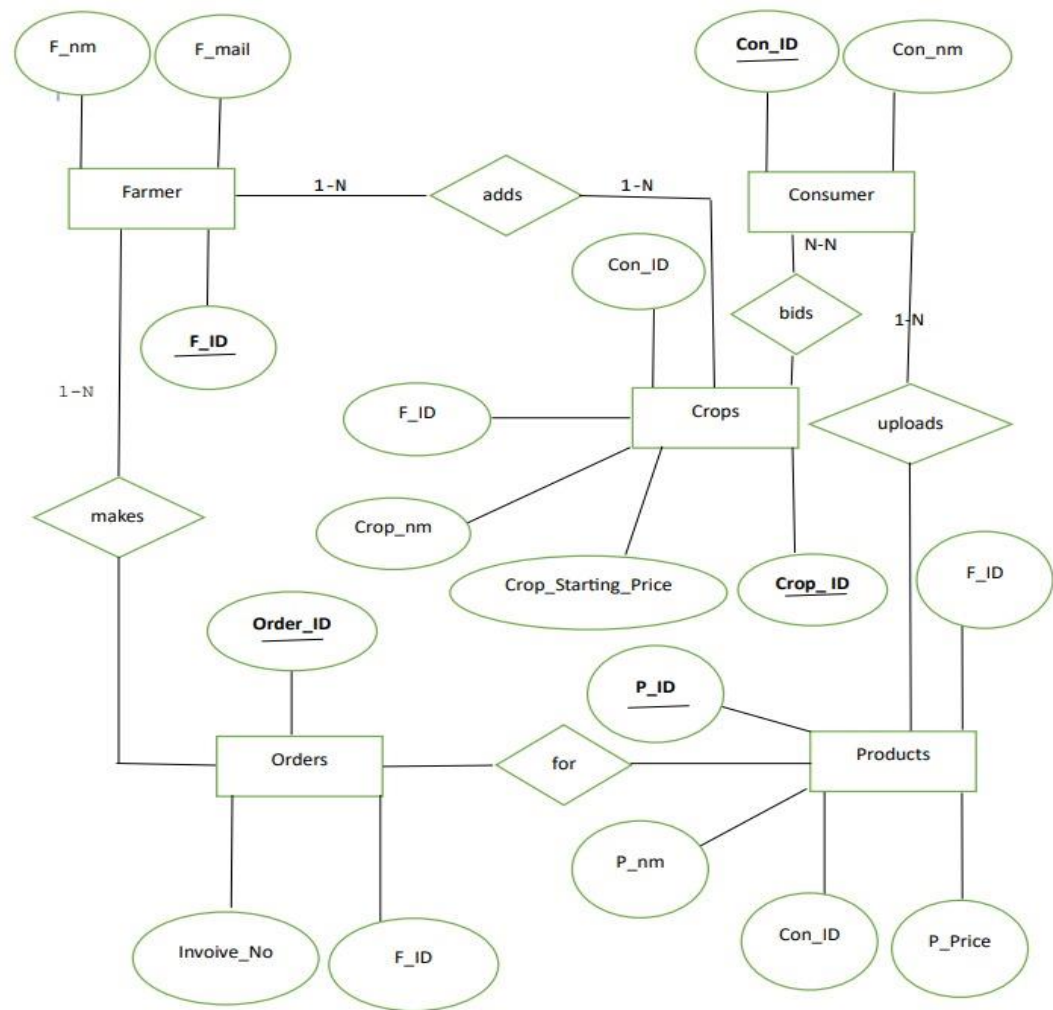


Figure 2.8 ER diagram

2.4. Detailed Design

2.4.1. Modular decomposition of the System

2.4.1.1. Functional Component 1: Admin

2.4.1.1.1. Design assumptions

This module is designed in such a way that it allows the administrator to use any of the options easily so that he can add, delete, modify or update any details. Admin module enables the admin to have control over the users and service providers.

2.4.1.2. Functional Component 1: Farmer

2.4.1.2.1. Design assumptions

This module is designed with an intention to allow Farmer to login, maintain their account, search for crops and update the product details. Farmer module is designed in such a way that it will be user friendly.

2.4.1.3. Functional Component 1: Consumer

2.4.1.3.1. Design assumptions

This module is designed to enable the consumer to create and maintain their account and input required details. This module will provide details of crops and the facility to login to their account. Consumer will be able to input or update data easily by filling in the given form.

2.4.2. Structure chart showing the hierarchy of modules

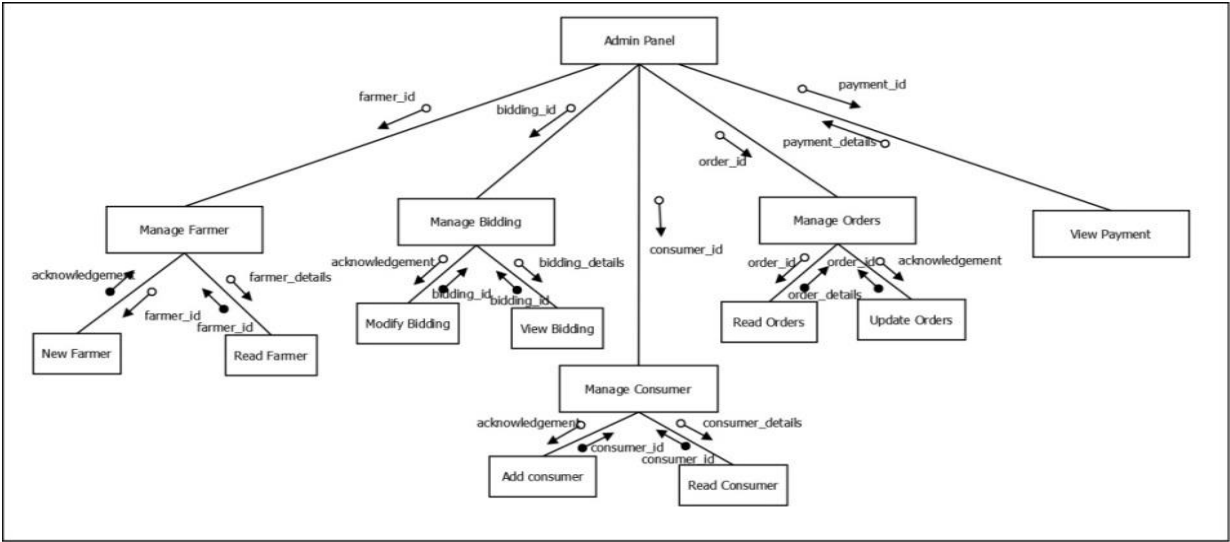


Figure 2.9 Structure chart of the Admin Login

2.4.2.1. Data Structures shared among modules

- Farmer
- Bidding
- Orders
- Payment

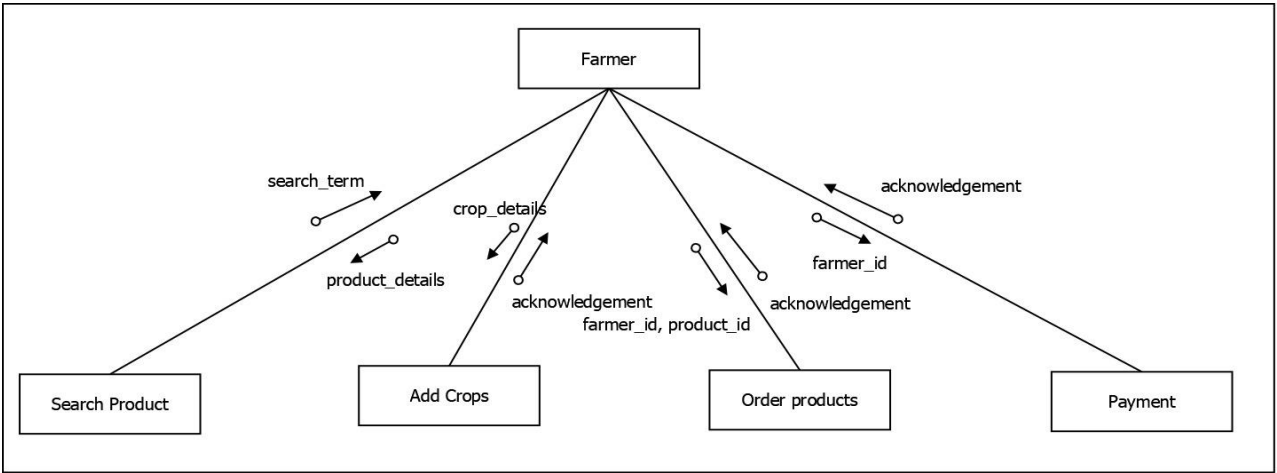


Figure 2.10 Structure chart of the Farmer

2.4.2.2. Data Structures shared among modules

- Product
- Crops
- Payment

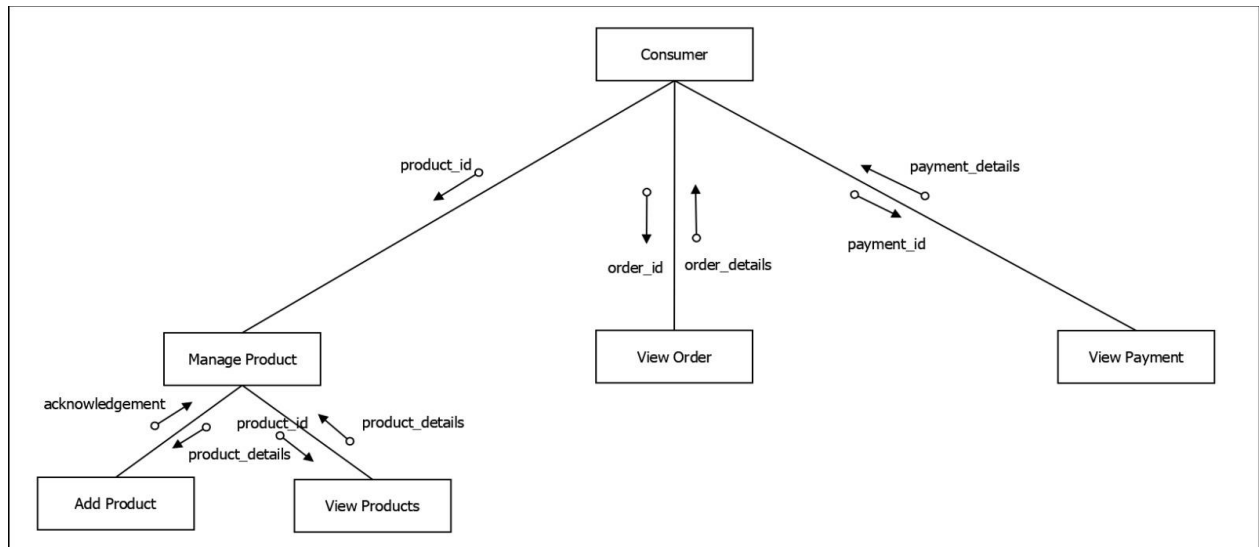


Figure 2.11 Structure chart of the consumer

2.4.2.3. Data Structures shared among modules

- Product
- Crops
- Payment

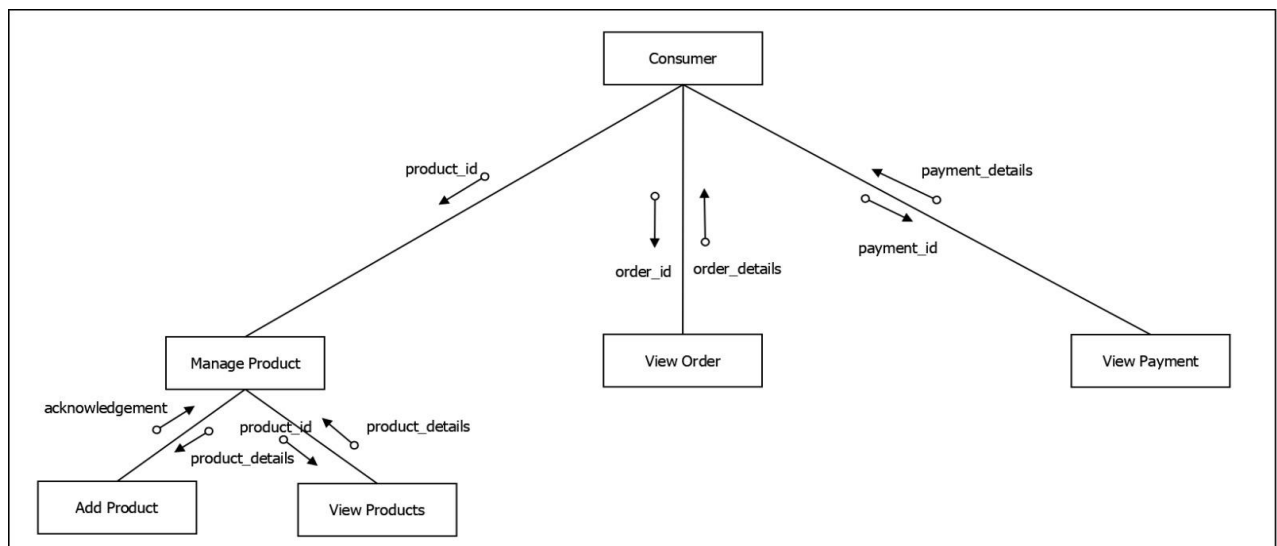


Figure 2.11 Structure chart of the consumer

2.4.2.4. Data Structures shared among modules

- Product
- Order
- Payment

2.4.3. Procedural details (Flow Chart) of components

Consumer

Inputs: cosumer_id, username

Procedural details (Flow Chart)

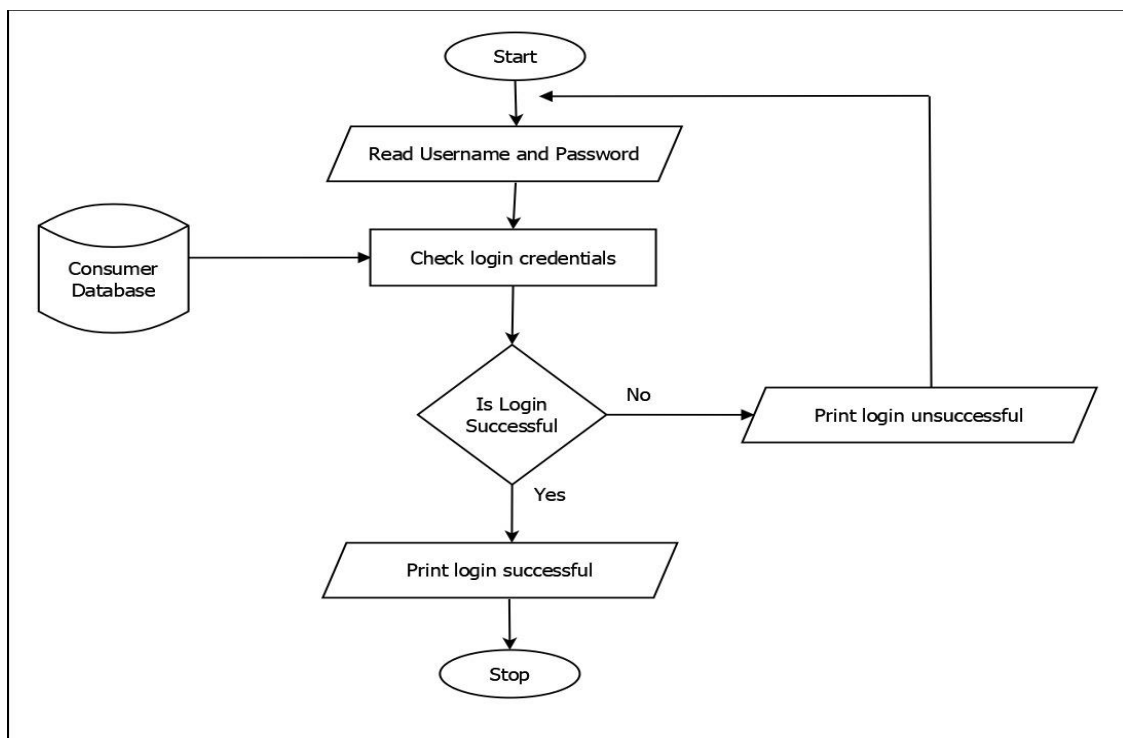


Figure 2.12 flow chart of the consumer

File I/O interfaces

The consumer is provided to add the details

Outputs: consumer details

View/add Consumer

Inputs: cosumer_id,view_consumer,add_consumer

Procedural details (Flow Chart)

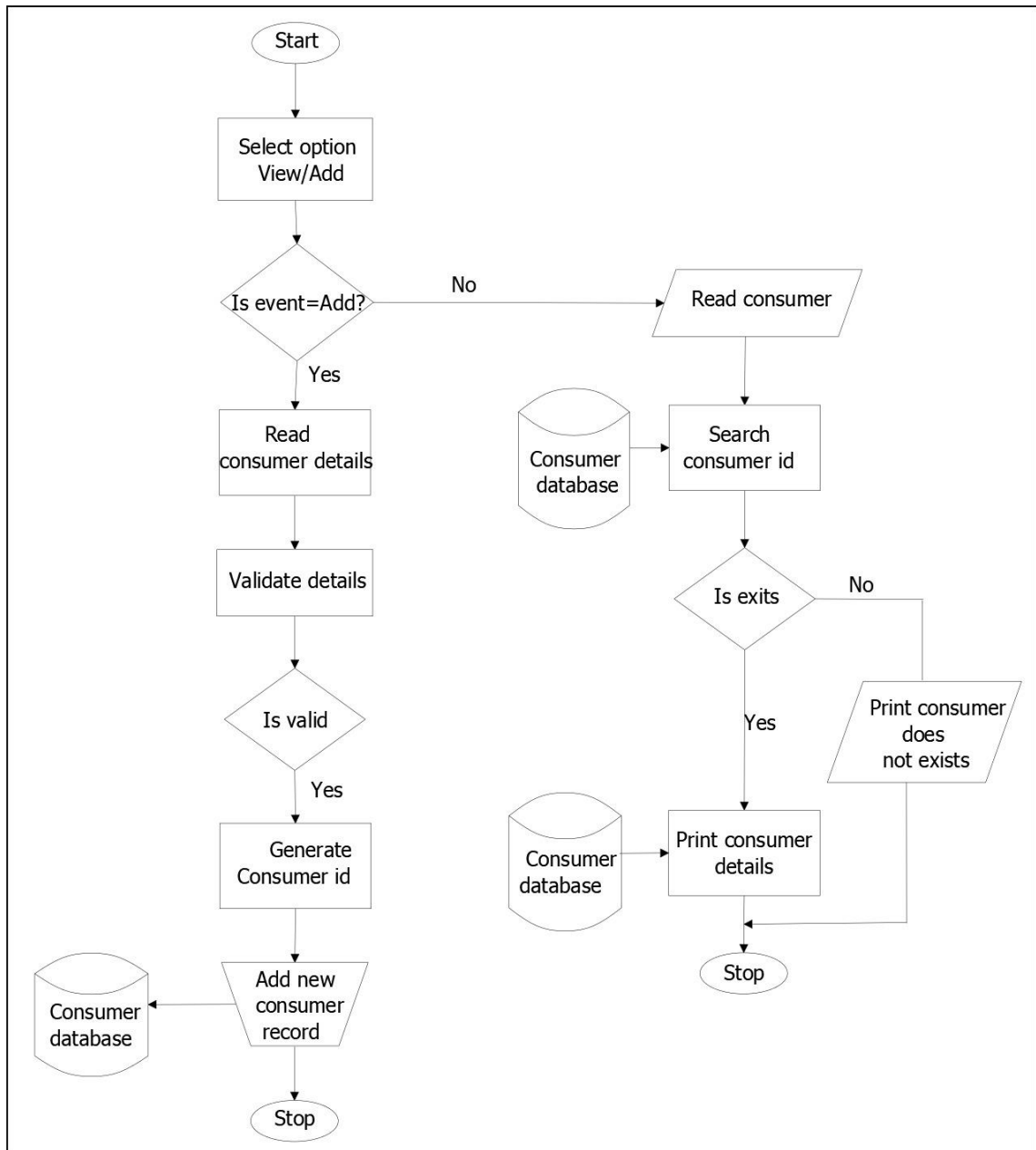


Figure 2.13 flow chart of the View/Add consumer

File I/O interfaces

The farmer can view the details of customers

Outputs: Adding of consumer, Viewing of consumer

View/Add Product

Inputs: product_id,view_product,add_product

Procedural details (Flow Chart)

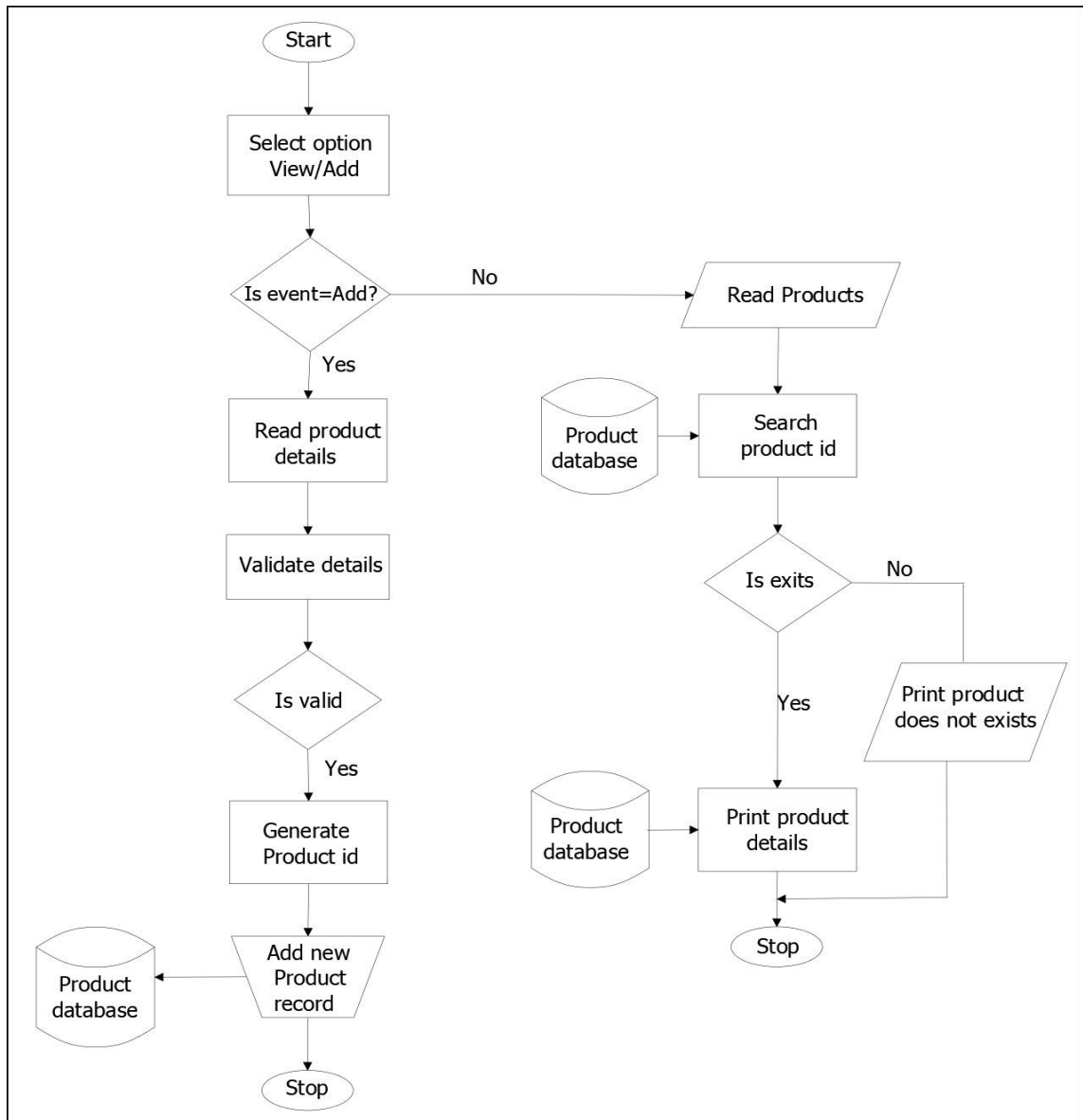


Figure 2.14 flow chart of the Product

File I/O interfaces

The farmer can add and view the product details

Outputs: add/view product details

View/Add Bidding

Inputs: bidding_id, view_bidding, add_bidding

Procedural details (Flow Chart)

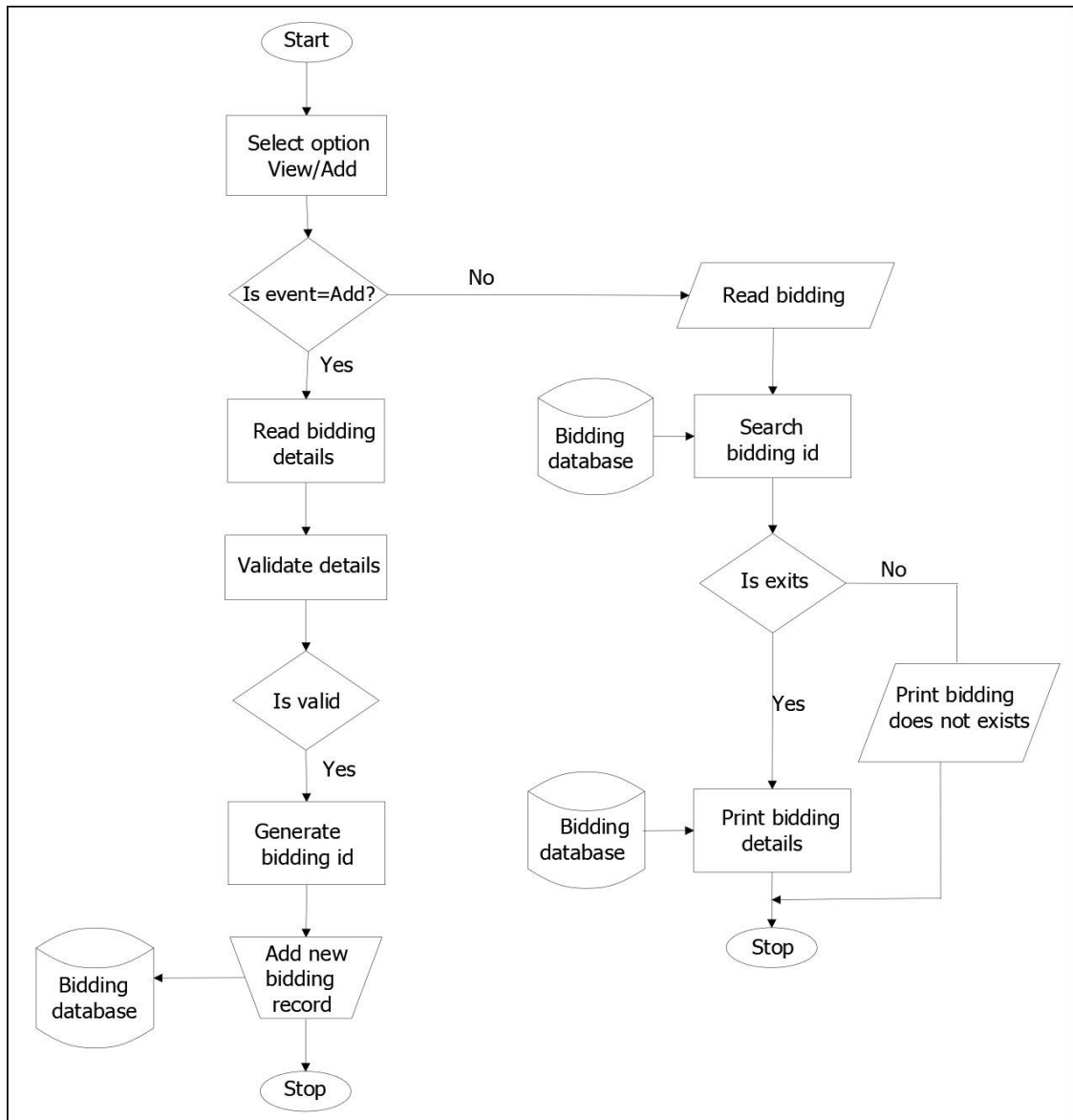


Figure 2.15 flow chart of the Bidding

File I/O interfaces

They can add and view the bidding and read the bidding done by the farmer

Outputs: add/view bidding details

Order

Inputs: farmer_id,product_id

Procedural details (Flow Chart)

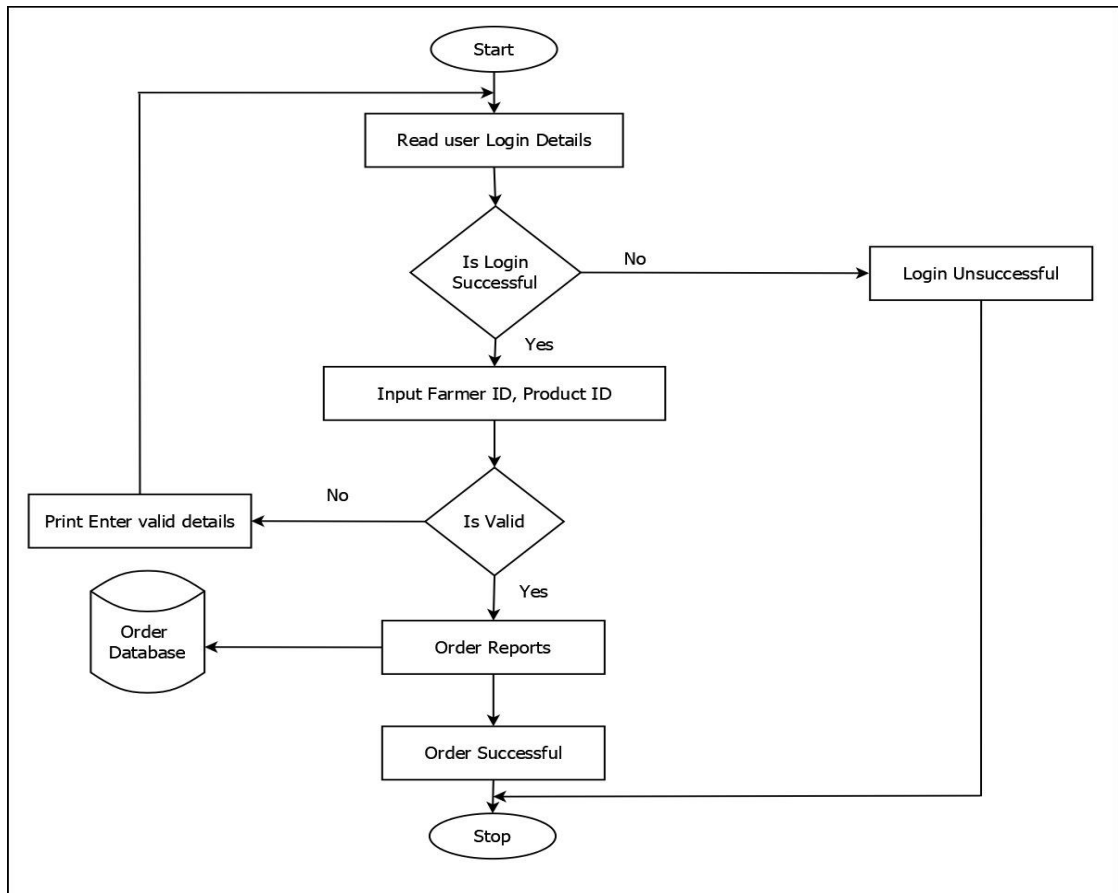


Figure 2.16 flow chart of the Order

File I/O interfaces

The user can log into the system, and they can view the order details

Output: order details

Order

Inputs: order_id

Procedural details (Flow Chart)

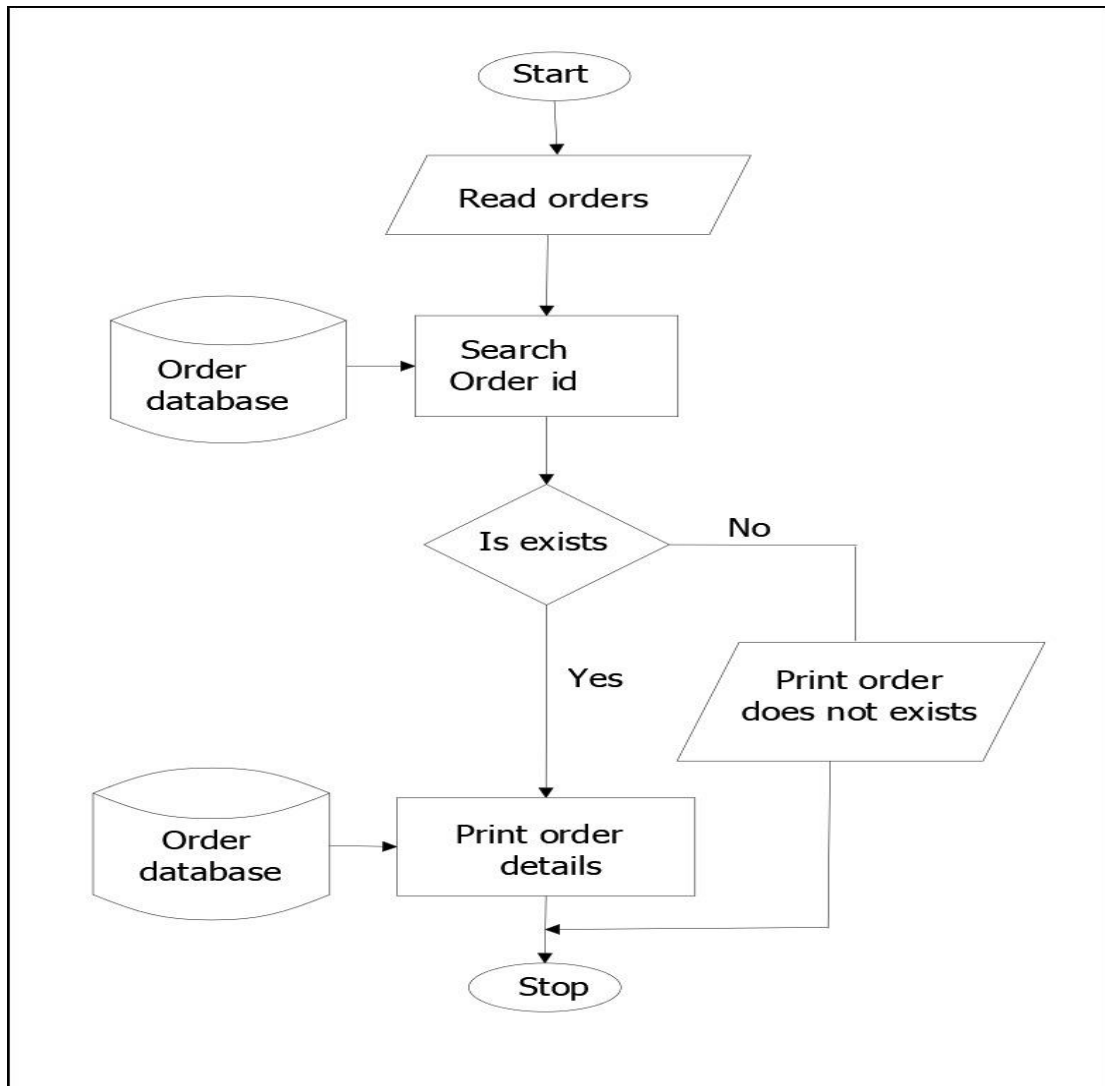


Figure 2.17 flow chart of the Search Order

File I/O interfaces

To input the order of crops

Output: order details

Product

Inputs: product_id

Procedural details (Flow Chart)

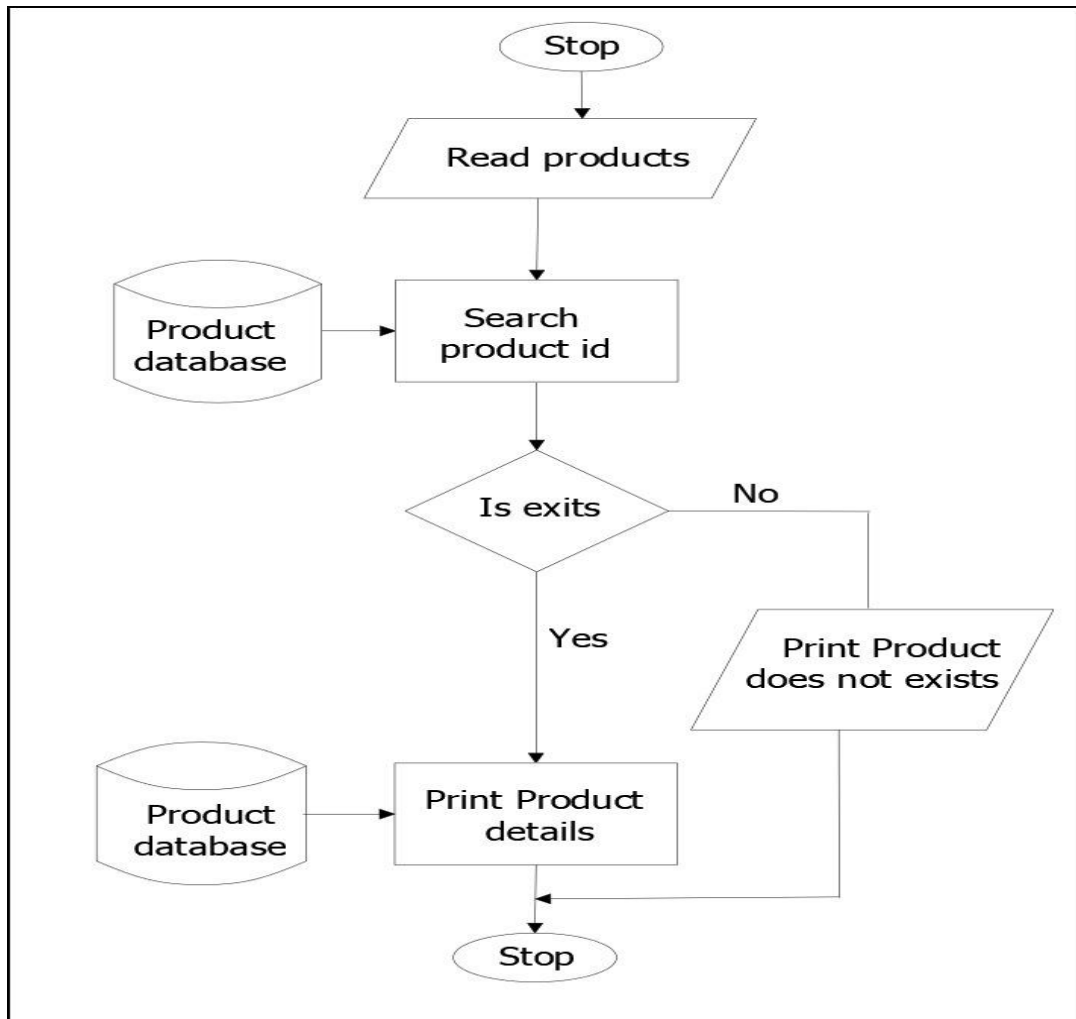


Figure 2.18 flow chart of the Product

File I/O interfaces

To input the order of crops

Outputs: product details

View/ Add Farmer

Inputs: famer_id,view_farmer, add_farmer

Procedural details (Flow Chart)

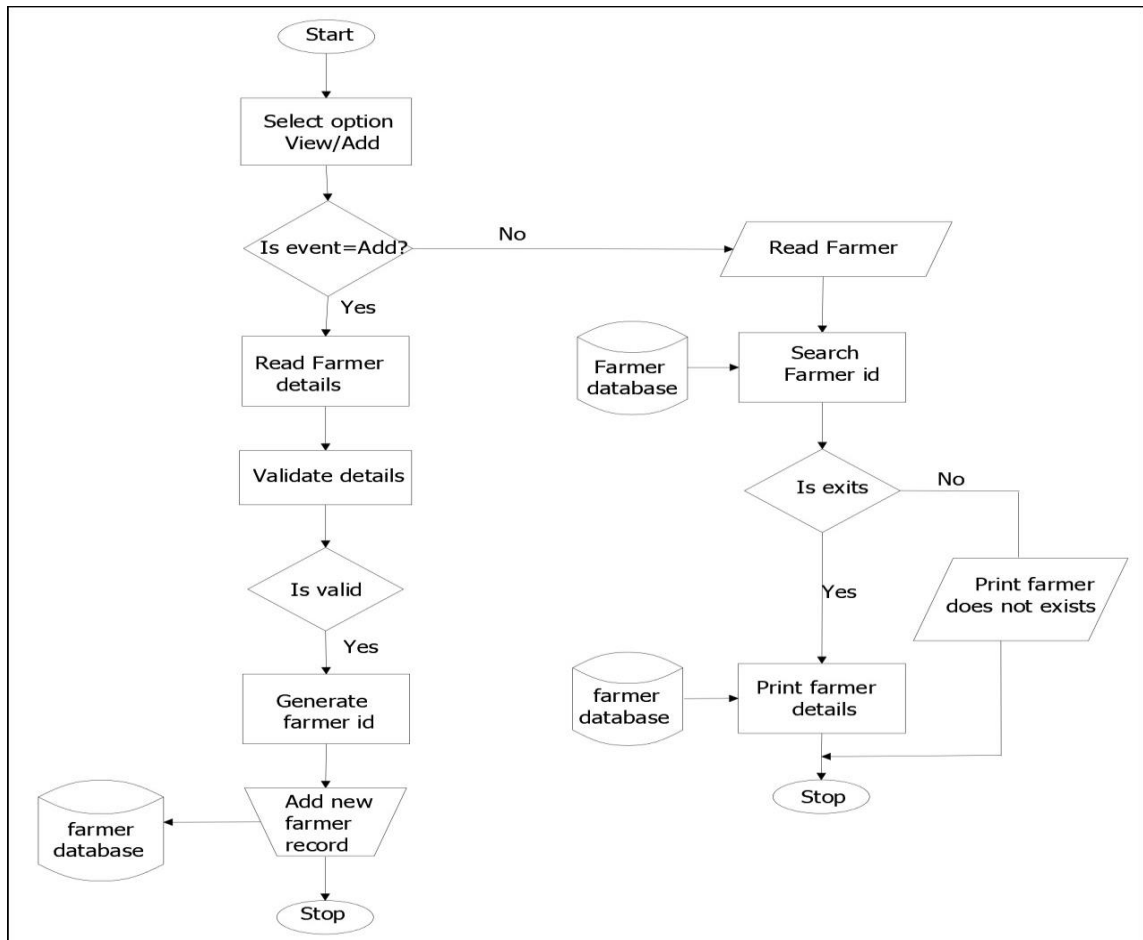


Figure 2.19 flow chart for the Farmer

File I/O interfaces

To input the details of farmer

Outputs: farmer details

Chapter 3

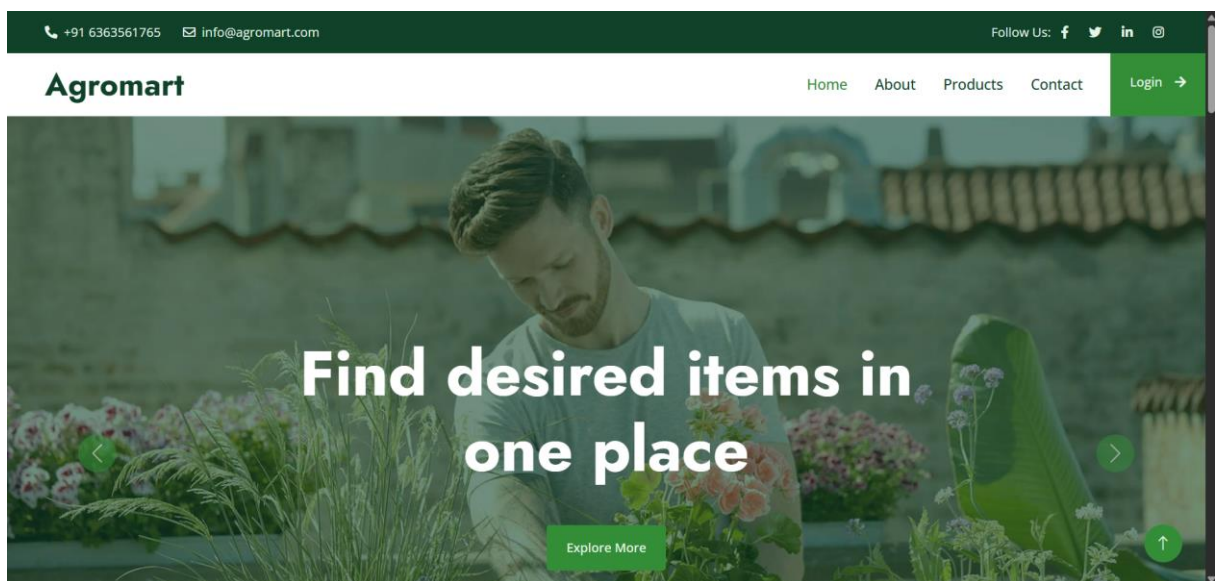
INPUT AND OUTPUT SCREENS

3.1. Introduction

The Web Application “Agromart”, is a user-friendly GUI based software which is developed using PHP as frontend tool and MySQL Server as backend. The system was developed such that the user has minimal number of inputs.

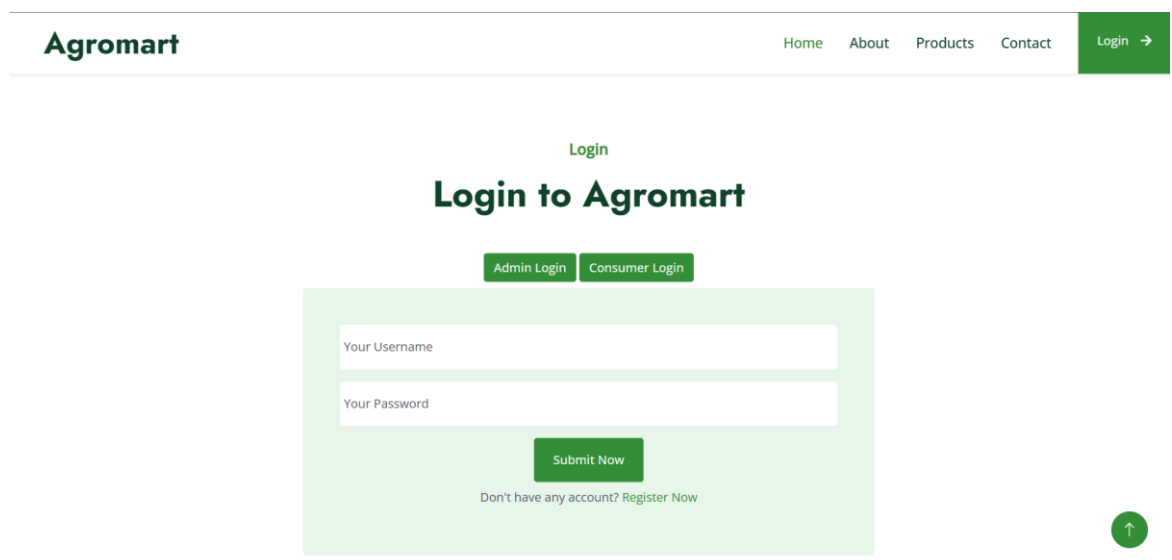
The users of this Web Application are Admin, Service Providers and Users. Admin is given complete access to all the functionality of the software.

3.2. Home Page



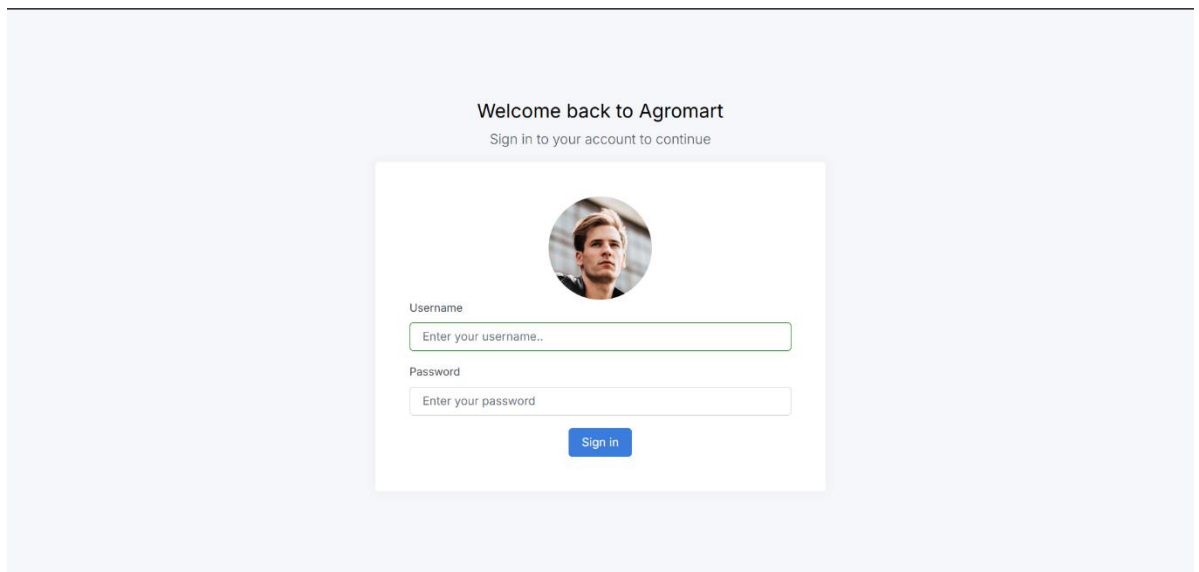
This is the homepage of Agromart; this page will be displayed as soon as the website link is clicked.

3.3. Login



This is the login screen for farmers. There are other links for Admin Login and Consumer Login.

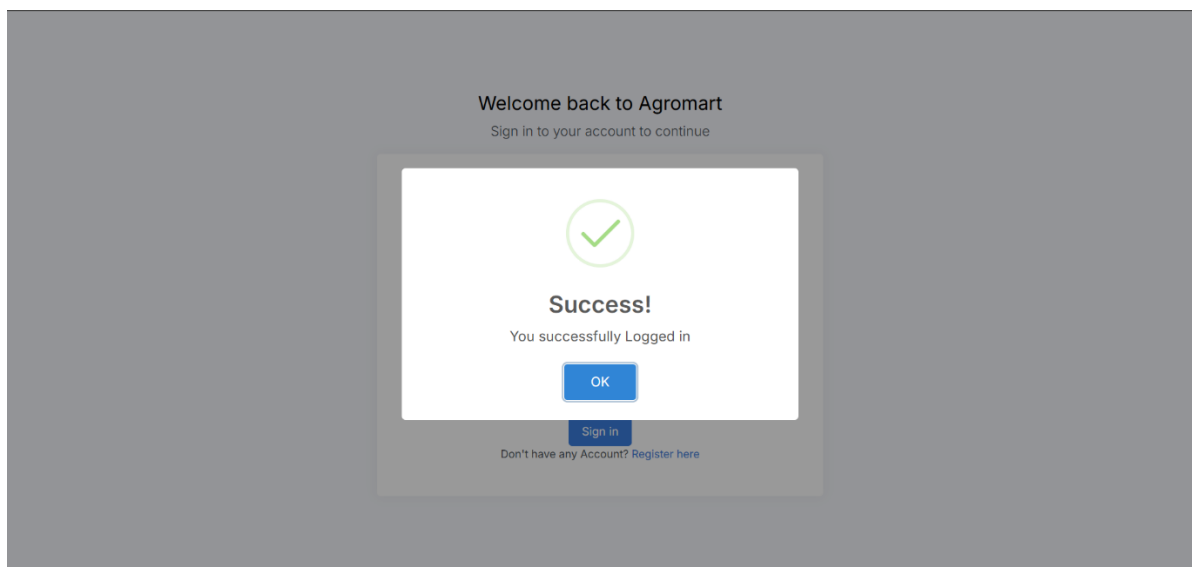
3.4. Admin Login



The image shows a login form for the 'Admin' role. The form is centered on a light blue background. At the top, it says 'Welcome back to Agromart' and 'Sign in to your account to continue'. Below this is a circular profile picture of a man. Underneath the picture are two input fields: 'Username' with the placeholder 'Enter your username..' and 'Password' with the placeholder 'Enter your password'. A blue 'Sign in' button is located below the password field.

This is the admin login page for admin. The admin manages the entire system. By entering the username and password the admin can login to the system.

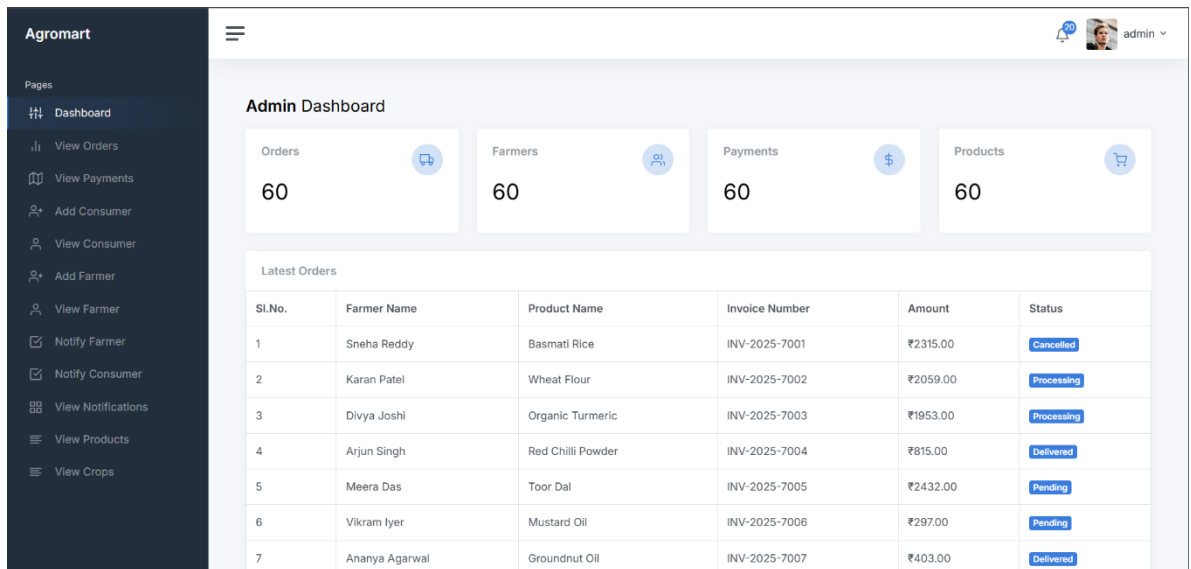
3.5. Successful Login



The image shows a confirmation screen for a successful login. The background is a solid grey. In the center, there is a white box with a green checkmark icon at the top. Below the icon, it says 'Success!' and 'You successfully Logged in'. There is a blue 'OK' button below the text. At the bottom of the white box, there is a blue 'Sign in' button and a link that says 'Don't have any Account? Register here'.

This is the page appears when the login is successful.

3.6. Admin Home

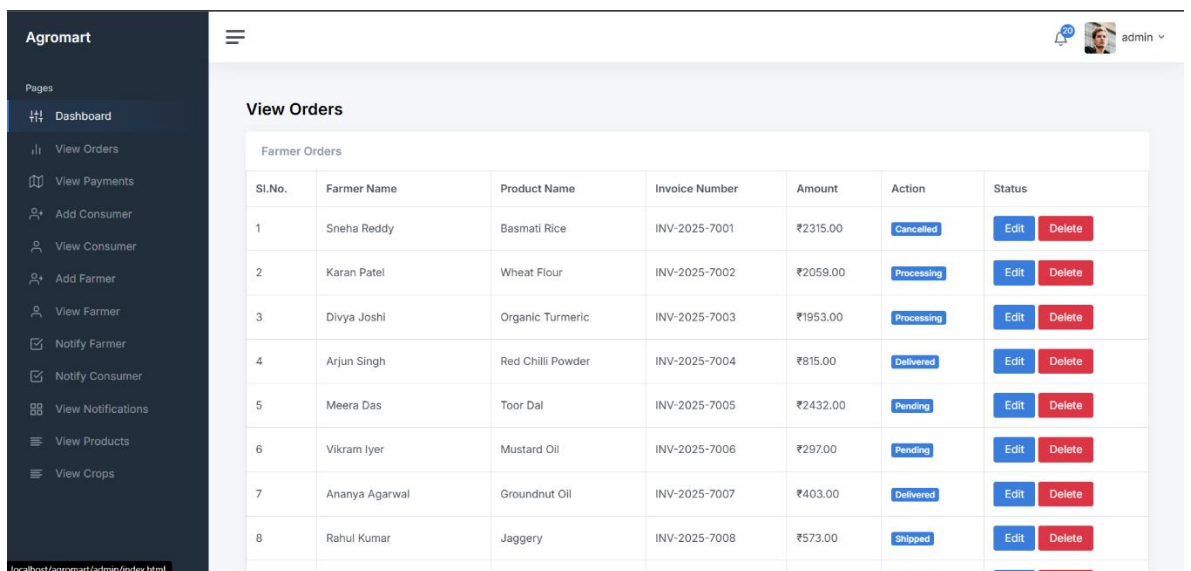


The screenshot shows the Agromart Admin Dashboard. On the left is a dark sidebar with the 'Agromart' logo and a 'Pages' menu. The 'Dashboard' page is selected. The main content area is titled 'Admin Dashboard' and features four summary cards: 'Orders' (60), 'Farmers' (60), 'Payments' (60), and 'Products' (60). Below these cards is a table titled 'Latest Orders' with 7 rows of order data. Each row includes a status button (Cancelled, Processing, or Delivered).

Sl.No.	Farmer Name	Product Name	Invoice Number	Amount	Status
1	Sneha Reddy	Basmati Rice	INV-2025-7001	₹2315.00	Cancelled
2	Karan Patel	Wheat Flour	INV-2025-7002	₹2059.00	Processing
3	Divya Joshi	Organic Turmeric	INV-2025-7003	₹1953.00	Processing
4	Arjun Singh	Red Chilli Powder	INV-2025-7004	₹815.00	Delivered
5	Meera Das	Toor Dal	INV-2025-7005	₹2432.00	Pending
6	Vikram Iyer	Mustard Oil	INV-2025-7006	₹297.00	Pending
7	Ananya Agarwal	Groundnut Oil	INV-2025-7007	₹403.00	Delivered

This is the home screen for admin. It contains the latest orders in the dashboard and other links to navigate to respected pages.

3.7. View orders

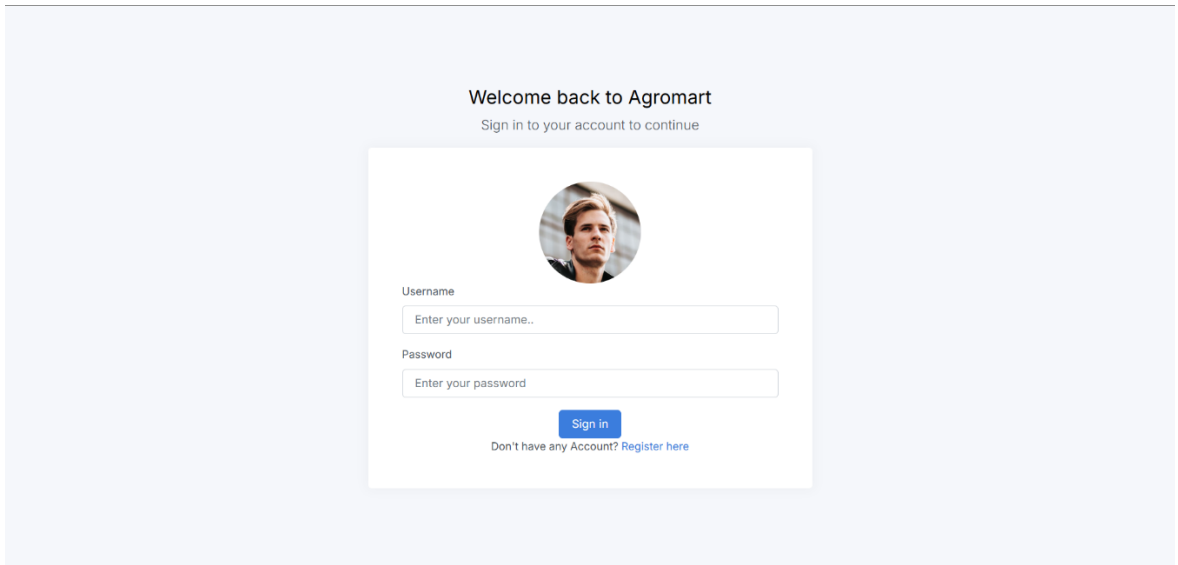


The screenshot shows the 'View Orders' page in the Agromart admin interface. It features a table titled 'Farmer Orders' with 8 rows. Each row includes an 'Action' column with a status button (Cancelled, Processing, Delivered, or Shipped) and a 'Status' column with 'Edit' and 'Delete' buttons. The sidebar on the left is identical to the previous screenshot, with 'View Orders' selected in the menu.

Sl.No.	Farmer Name	Product Name	Invoice Number	Amount	Action	Status
1	Sneha Reddy	Basmati Rice	INV-2025-7001	₹2315.00	Cancelled	Edit Delete
2	Karan Patel	Wheat Flour	INV-2025-7002	₹2059.00	Processing	Edit Delete
3	Divya Joshi	Organic Turmeric	INV-2025-7003	₹1953.00	Processing	Edit Delete
4	Arjun Singh	Red Chilli Powder	INV-2025-7004	₹815.00	Delivered	Edit Delete
5	Meera Das	Toor Dal	INV-2025-7005	₹2432.00	Pending	Edit Delete
6	Vikram Iyer	Mustard Oil	INV-2025-7006	₹297.00	Pending	Edit Delete
7	Ananya Agarwal	Groundnut Oil	INV-2025-7007	₹403.00	Delivered	Edit Delete
8	Rahul Kumar	Jaggery	INV-2025-7008	₹573.00	Shipped	Edit Delete

This is view orders page where admin can view all the orders and manage them by editing or deleting.

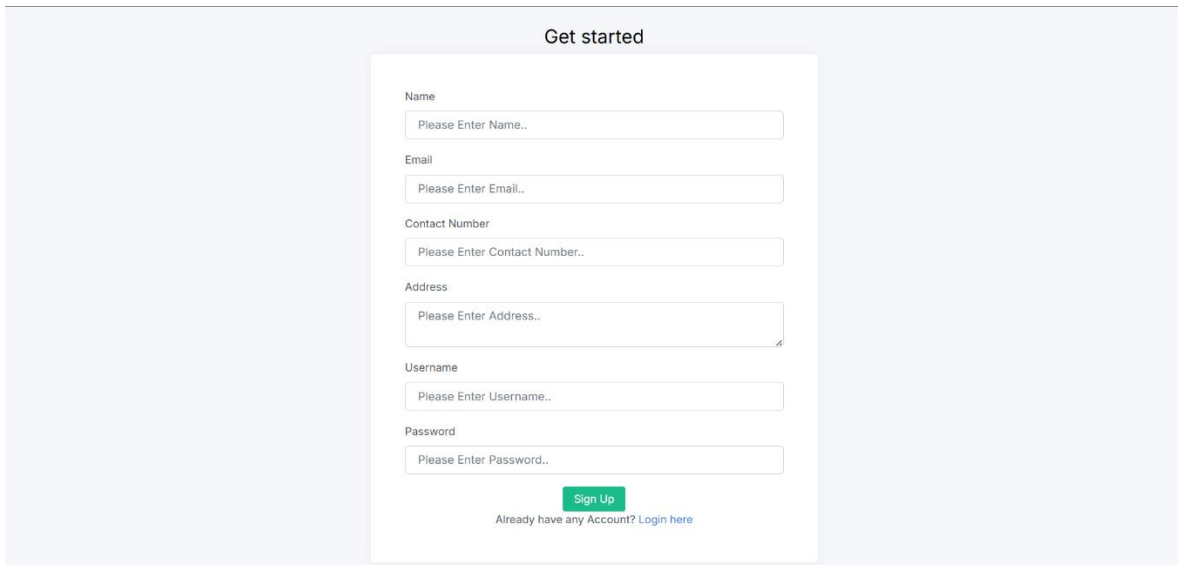
3.8. Consumer Login



The image shows a consumer login page for 'Agromart'. At the top, it says 'Welcome back to Agromart' and 'Sign in to your account to continue'. Below this is a white card containing a circular profile picture of a man. Under the picture are two input fields: 'Username' with the placeholder 'Enter your username..' and 'Password' with the placeholder 'Enter your password'. A blue 'Sign in' button is positioned below the password field. At the bottom of the card, there is a link that says 'Don't have any Account? Register here'.

This is consumer login page.

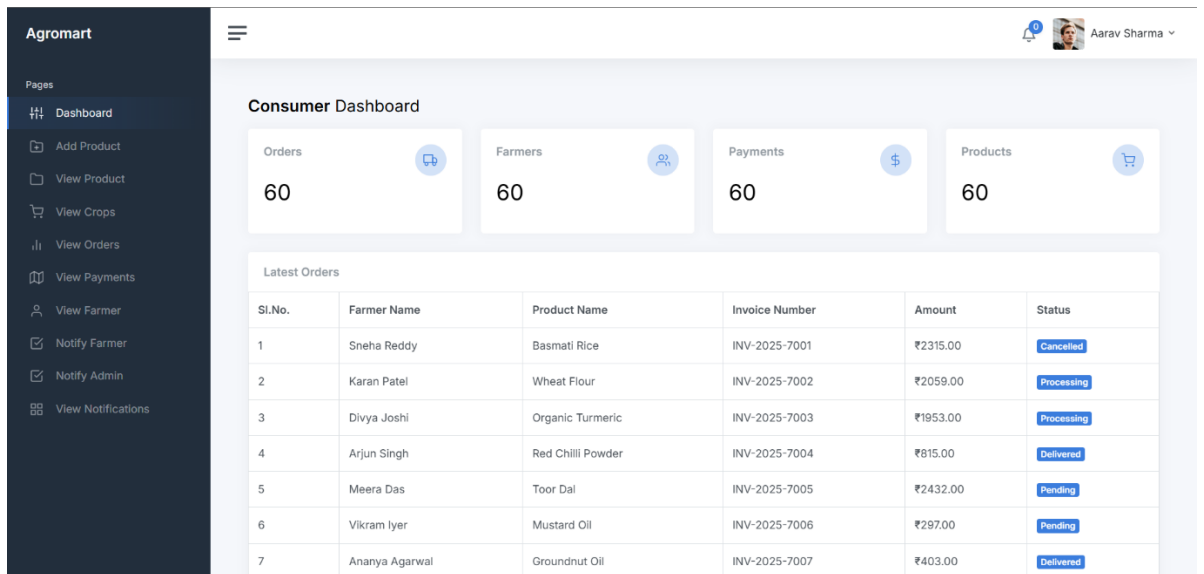
3.9. Consumer Registration



The image shows a consumer registration page for 'Agromart'. At the top, it says 'Get started'. Below this is a white card with several input fields: 'Name' (placeholder: 'Please Enter Name..'), 'Email' (placeholder: 'Please Enter Email..'), 'Contact Number' (placeholder: 'Please Enter Contact Number..'), 'Address' (placeholder: 'Please Enter Address..'), 'Username' (placeholder: 'Please Enter Username..'), and 'Password' (placeholder: 'Please Enter Password..'). A green 'Sign Up' button is located below the password field. At the bottom of the card, there is a link that says 'Already have any Account? Login here'.

This is consumer registration page.

3.10. Consumer Home



Agromart

Pages

- Dashboard
- Add Product
- View Product
- View Crops
- View Orders
- View Payments
- View Farmer
- Notify Farmer
- Notify Admin
- View Notifications

Consumer Dashboard

Orders: 60

Farmers: 60

Payments: 60

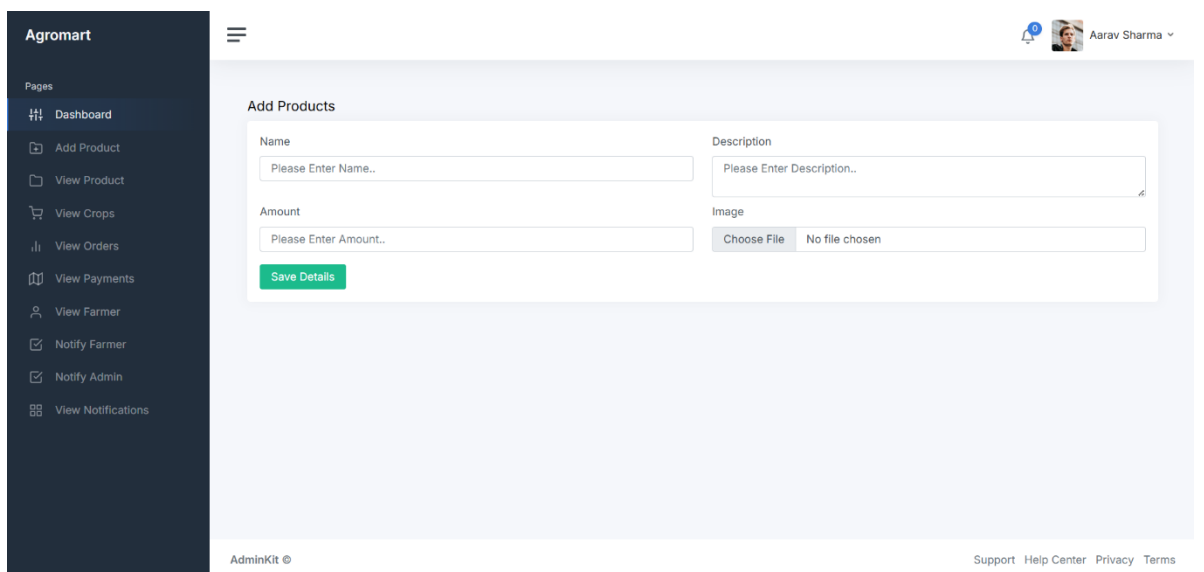
Products: 60

Latest Orders

Sl.No.	Farmer Name	Product Name	Invoice Number	Amount	Status
1	Sneha Reddy	Basmati Rice	INV-2025-7001	₹2315.00	Cancelled
2	Karan Patel	Wheat Flour	INV-2025-7002	₹2059.00	Processing
3	Divya Joshi	Organic Turmeric	INV-2025-7003	₹1953.00	Processing
4	Arjun Singh	Red Chilli Powder	INV-2025-7004	₹815.00	Delivered
5	Meera Das	Toor Dal	INV-2025-7005	₹2432.00	Pending
6	Vikram Iyer	Mustard Oil	INV-2025-7006	₹297.00	Pending
7	Ananya Agarwal	Groundnut Oil	INV-2025-7007	₹403.00	Delivered

This is consumer home page and he has the latest orders in dashboard and other links which related to him.

3.11. Add Product Page



Agromart

Pages

- Dashboard
- Add Product
- View Product
- View Crops
- View Orders
- View Payments
- View Farmer
- Notify Farmer
- Notify Admin
- View Notifications

Add Products

Name:

Description:

Amount:

Image: No file chosen

AdminKit © Support Help Center Privacy Terms

This is add product page from where the consumer adds the product to the system.

3.12. Farmer Home

Agromart

Pages

- Home
- Dashboard**
- Add Crop
- View Crop
- View Products
- View Orders
- View Payments
- View Consumer
- Notify Consumer
- Notify Admin
- View Notifications

Admin Dashboard

Orders: 1 Farmers: 60 Payments: 1 Products: 60

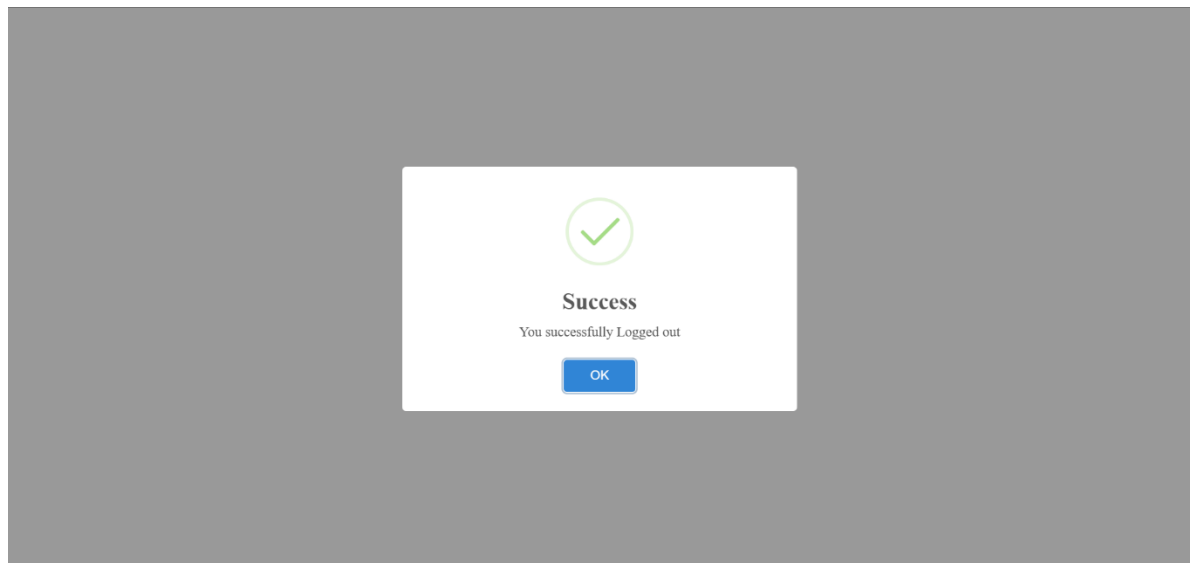
Latest Orders

Sl.No.	Farmer Name	Product Name	Invoice Number	Amount	Status
1	Sneha Reddy	Basmati Rice	INV-2025-7001	₹2315.00	Cancelled

AdminKit © Support Help Center Privacy Terms

This is farmer's home page and he has the links which is related to him.

3.13. Logout



This is logout screen after the user is logged out of the system.

Chapter 4

Testing

4.1. Introduction

Software testing is an investigation conducted to provide stakeholders with information about the quality of the product or service under test. Testing has been defined as the process of analyzing a software item to detect the differences between existing and required conditions and to evaluate the features of the software item. Software testing is the process used to assess the quality of computer software.

It involves the operation of a system or application under controlled conditions and evaluating the results. The conditions controlled should include both normal and abnormal conditions. Testing should intentionally attempt to make things go wrong to determine things happen when they should. It is oriented to 'detection'.

Software testing has three main purposes:

- The verification process confirms that the software meets its technical specifications. A "specification" is a description of a function in terms of a measurable output value given a specific input value under specific preconditions.
- The validation process confirms that the software meets the business requirements.
- A defect is a variance between the expected and actual result. The defect's ultimate source may be traced to a fault introduced in the specification, design or development phases. Not all the defects with which necessarily result in failure.

There are two types of software testing:

- Black box testing -internal system design - is not considered in this type of testing. Tests are based on requirements and functionality.
- White box testing-this testing is based on knowledge of the internal logic of an application's code. Also known as glass box testing. Tests are based on coverage of code statements, branches, paths and conditions.

A test case is a software testing document, which consists of events, actions, input, output, expected result and actual result. Clinically defined a test case is an input and an expected result. This can be pragmatic as x your derived result is y '; whereas other test cases describe in more detail the input scenario and what results might be expected outcome. It can occasionally be a series of steps but one with expected results or expected outcome. A test case should also contain a place for the actual result.

White box testing is applicable at the unit, integration and system levels of software testing process.

4.2. Test Reports

4.2.1. Unit Testing

Unit testing focuses effort on the smallest unit of software design. This is known as module testing. The modules are tested separately. The test is carried out during programming stage itself. In this step, each module is found to be working satisfactory as regards the expected output from the module.

Test cases

4.2.1.1. Admin Login form

Test Scenario ID		Login-1		Test Case ID			Login-1A
Test Description Case		Login – successful or not		Test Priority			High
Pre-Requisite		A valid user account		Post-Requisite			NA
Test Execution Steps:							
S. No	Action	Inputs	Expected Output	Actual Output	Test Browser	Test Result	Test Comments
1	If user clicks on login button without entering username and password.	Inputs are not. given	Please enter username and password	Please enter username and password	Chrome	Pass	[26/04/2026 11:44 AM]: invalid login attempt
2	Enter correct username & Password and hit login button	username test@xyz.com Password: *****	Login success	Login success	Chrome	Pass	[26/04/2026 11:44 AM]: Login successful
3	If username is blank but password is entered.	Password: *****	Please enter username	Please enter username	Chrome	Pass	[26/04/2026 11:44 AM]: invalid login attempt
4	If password is blank but username is	Username: Mokshith	Please enter password	Please enter password	Chrome	Pass	[26/04/2026 11:44 AM]: invalid login attempt

	entered.						
5	If the username or Password is incorrect.	Username: Mokshith Password: *****	Invalid login credentials	Invalid login credentials	Chrome	Pass	[26/04/2026 11:44 AM]: Invalid login attempt

4.2.1.2. User Login form

Test Scenario ID	Login-1	Test Case ID	Login-1A
Test Description Case	Login – successful or not	Test Priority	High
Pre-Requisite	A valid user account	Post-Requisite	NA

Test Execution Steps:

S. No	Action	Inputs	Expected Output	Actual Output	Test Browser	Test Result	Test Comments
1	If user Click on login button without entering username and password.	Inputs are not given	Please enter username and password	Please enter username and password	Chrome	Pass	[26/04/2026 11:44 AM]: invalid login attempt
2	Enter correct username & Password and hit login. button	username test@xyz.com Password: *****	Login success	Login success	Chrome	Pass	[26/04/2026 11:44 AM]: Login successful

3	If username is blank but password is entered.	Password: *****	Please enter username	Please enter username	Chrome	Pass	[26/04/2026 11:44 AM]: Invalid login attempt
4	If password is blank but username is entered.	Username: Mathew	Please enter password	Please enter password	Chrome	Pass	[26/04/2026 11:44 AM]: Invalid login attempt
5	If the username or password is incorrect.	Username :Mokshith Password: *****	Invalid login credentials	Invalid login credentials	Chrome	Pass	[26/04/2026 11:44 AM]: invalid login attempt

4.2.2. Integrate Testing

Data can be lost across an interface. One module can have an adverse effect on another, sub functions, when combined, may not be linked in desired manner in major functions. Integration testing is a systematic approach for constructing the program structure, while at the same time conducting tests to uncover errors associated within the interface. The objective is to take unit tested modules and build program structure. All the modules are combined and tested as a whole.

4.2.3. System Testing

System testing refers to the testing of complete system that is carried out by testers sometimes by the developers to check whether it meets the system specifications or not. System testing involves both functional and non-functional testing. In the system the behavior of whole system product is tested as defined by the scope of the development project. It may include tests based on risks or requirement specification, business process, use cases or other high-level description of the system behavior, interaction with the operating system, and system resources.

SL.NO	Test Condition	Test Result
1	System Loading	Successful
2	System Run Procedure	Successful
3	File I/O Operation	Successful
4	Database Communication	Successful
5	Server/Client Interaction	Successful
6	Memory Usage	Normal
7	System Processor Usage	Normal
8	Authentication/Authorization	Successful
9	Back Up/Recovery	Successful

Chapter 5

CONCLUSION

5.1. Limitations

Agromart has some drawbacks that we have observed:

- Currently it is only a web application, it is not available as an android or iOS application.
- The application now runs only in local system.
- The application is currently used with the few modules.
- The backend storage is limited.

5.2. Scope for Enhancement (Future scope)

- More modules can be added.
- Application can be hosted in server.
- In future, this web application could be integrated with the android and iOS applications.

5.3. Conclusion

Agromart is a web-based project that aims to help Farmers and Consumers. The main users of the system, the Farmers usually face a lot of problems in finding the good fertilizers, quality equipment's or any other agriculture related items that will be helpful for them. The proposed system Agromart will solve the problem by developing a system that will provide all the agriculture-related information in one place.

5.4. Abbreviation and Acronyms (Lists)

I/O: Input/Output

OS: Operating System DFD: Data Flow Diagram ADMIN: The Administrator

CFD: Context Flow Diagram

RDBMS: Relational Database Management System

RAM: Random Access Memory

SQL: Structured Query Language

SRS: Software Requirement Specification

IIS: Internet Information Server HTML5: Hyper Text Markup Language5 CSS: Cascading Style Sheet

ER Diagram: Entity Relationship Diagram

1NF: First Normal Form 2NF: Second Normal Form 3NF: Third Normal Form PK: Primary Key

FK: Foreign Key

DBMS: Database Management System

5.5. Bibliography/ References (list in specified format)

- <https://www.tutorialspoint.com/php/index.htm>
- <https://javascript.info/>
- <https://www.tutorialspoint.com/javascript/index.htm>
- <https://www.codecademy.com/learn/introduction-to-javascript>
- Learning PHP, MySQL, JavaScript, CSS, and HTML5: A Step-by-Step Guide to Creating Dynamic Website by Robin Nixon (Author)
- PANKAJ JALOTE's SOFTWARE ENGINEERING: A PRECISE APPROACH by Pankaj Jalote (Author)